



Batterjee Medical College
For Science and Technology

Ischemic Heart Disease

Approach to a Case of Chest Pain

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Learning Objectives

Knowledge

- Describe the epidemiology, clinical manifestations, and complications of coronary artery disease in Saudi Arabia and worldwide.
- Identify the major complications of myocardial infarction according to their expected time of onset post-infarction.

Skill

- Interpret ECG findings and cardiac biomarker patterns to differentiate STEMI from NSTEMI/unstable angina.
- Select the appropriate diagnostic test for a patient with suspected stable angina based on pretest probability of CAD.
- Formulate an evidence-based management plan IHD, including pharmacotherapy, secondary prevention, and indications for revascularization.

Value

- Demonstrate adherence to institutional and national (SCFHS/SMLE) protocols for time-critical STEMI management, including door-to-balloon/door-to-needle targets.

Main Subtopics

- Coronary Artery Anatomy
- Ischemic Heart disease
 - Definitions
 - Epidemiology
 - Etiology
 - Pathophysiology
 - Clinical Features
 - Diagnostics
 - Management
- Acute Coronary Syndrome
- MI Complications
- Subtypes and Variants of angina

Case Scenario

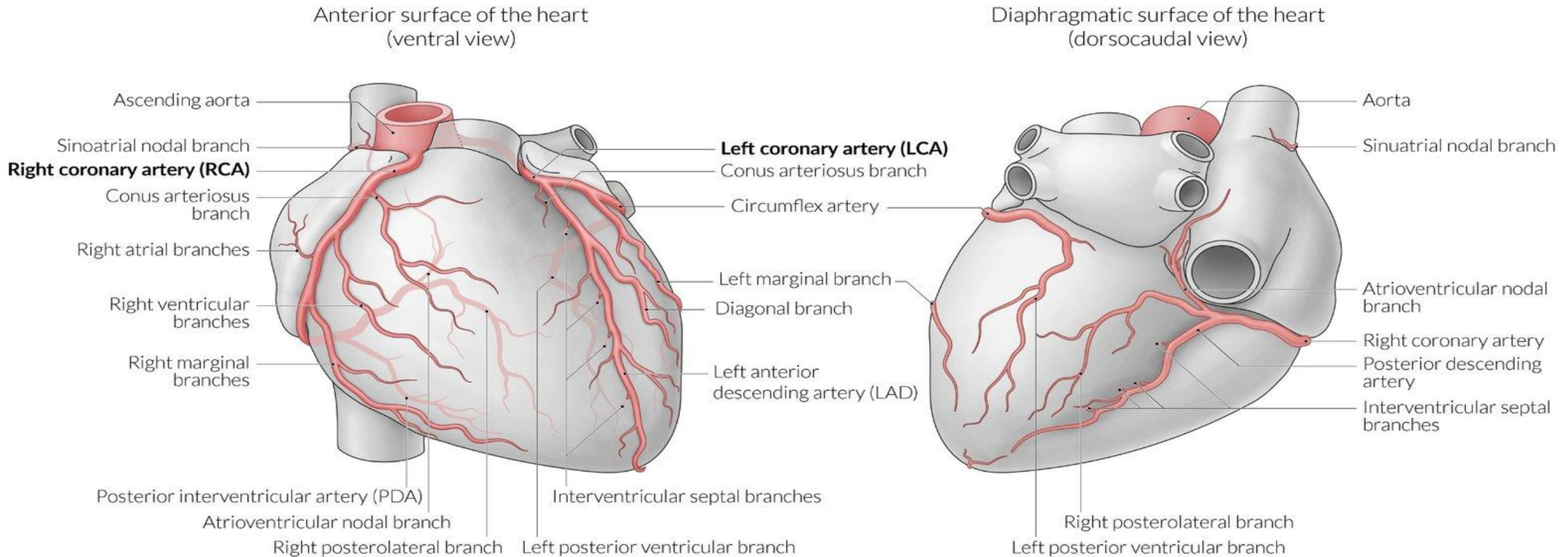
A 67-year-old man comes to the physician because of a 2-month history of chest pain. The pain is episodic and lasts for 5–10 minutes before it subsides spontaneously; it is localized to the left side of the chest and does not radiate. The patient reports that the pain typically occurs when he is angry or upset. He has not noticed any changes in the pattern, intensity, or duration of pain during these episodes. He does not report any other symptoms and there is no chest pain at the time of presentation. The patient has a history of asthma and hypertension. His medications are albuterol as needed and amlodipine. He goes swimming once a week and tries to maintain a healthy diet. He drinks one to two glasses of wine on weekends. He does not smoke or use illicit drugs. His temperature is 37.0°C (98.6°F), pulse is 88/min, respirations are 12/min, and blood pressure is 130/88 mm Hg. Cardiac examination shows a point of maximal impulse in the 5th intercostal space at the midclavicular line. Auscultation shows no murmurs, rubs, or gallops. An ECG shows a sinus rhythm and no abnormalities.

Case Scenario

At this visit, which of the following is the most appropriate next step to evaluate for coronary artery disease in this patient?

- a. No testing needed
- b. Troponin I measurement
- c. Stress myocardial perfusion imaging
- d. Cardiac catheterization
- e. Dipyridamole stress echocardiography
- f. Exercise Stress test

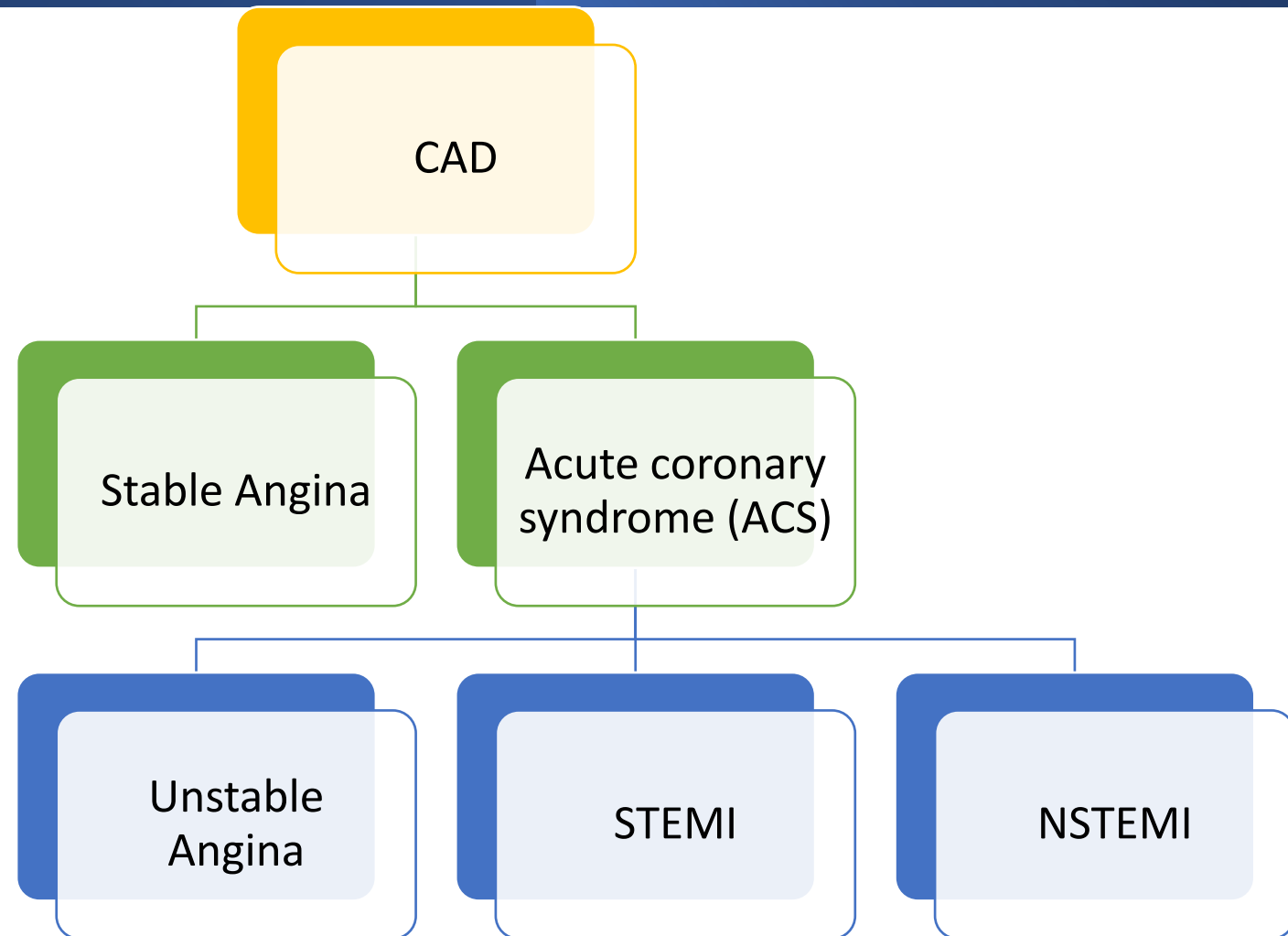
Coronary Artery Anatomy



Definition

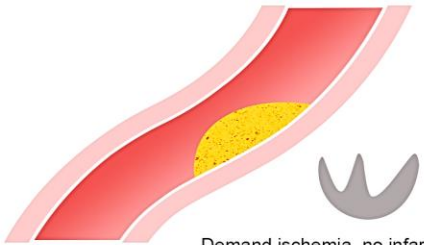
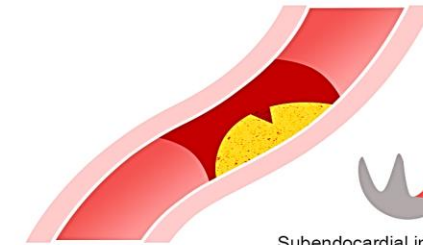
Coronary artery disease (CAD):

Ischemic heart disease due to narrowing or blockage of coronary arteries, resulting in a mismatch between myocardial oxygen supply and demand



Definition

- **Stable angina:** ↑ oxygen demand, ↓ supply
- **Unstable angina:** oxygen demand is unchanged, ↓ supply
- **Myocardial infarction (MI):** Necrosis of myocardium due to ↓ supply

ACUTE CORONARY SYNDROME	1 STABLE ANGINA	2 UNSTABLE ANGINA	3 NSTEMI	4 STEMI
	Angina pain develops when there is increased demand in the setting of a stable atherosclerotic plaque. The vessel is unable to dilate enough to allow adequate blood flow to meet the myocardial demand.	The plaque ruptures and a thrombus forms around the ruptured plaque, causing partial occlusion of the vessel. Angina pain occurs at rest or progresses rapidly over a short period of time.	During an NSTEMI, the plaque rupture and thrombus formation causes partial occlusion to the vessel that results in injury and infarct to the subendocardial myocardium.	A STEMI is characterized by complete occlusion of the blood vessel lumen, resulting in transmural injury and infarct to the myocardium, which is reflected by ECG changes and a rise in troponins.
	 Demand ischemia, no infarct	 Supply ischemia, no infarct	 Subendocardial infarct	 Transmural infarct
	 Normal	 Normal, Inverted T waves, or ST depression	 Normal, Inverted T waves, or ST depression	 Hyperacute T waves or ST elevation
TROPONINS	Normal	Normal	Elevated	Elevated

Definition

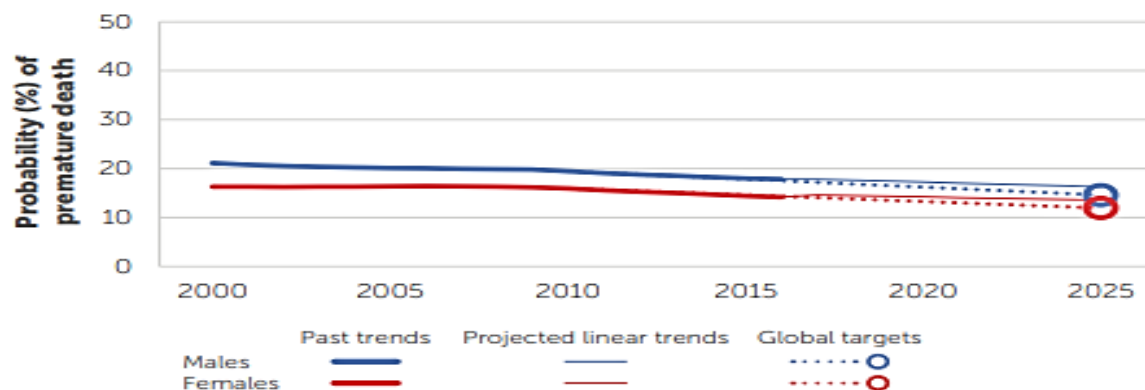
Typical angina (definite)	Meets all three of the following characteristics: • substernal chest discomfort of characteristic quality and duration; • provoked by exertion or emotional stress; • relieved by rest and/or nitrates within minutes.
Atypical angina (probable)	Meets two of these characteristics.
Non-anginal chest pain	Lacks or meets only one or none of the characteristics.

Epidemiology

- CAD is the leading cause of death all over the world
- Cardiovascular disease accounted for 37% of all deaths in Saudi in 2018
- According to one study, overall prevalence of CAD in KSA is 5.5%
- The prevalence in males and females were 6.6% and 4.4%

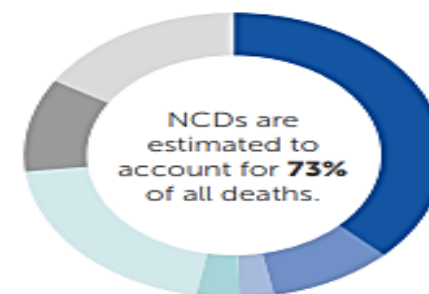
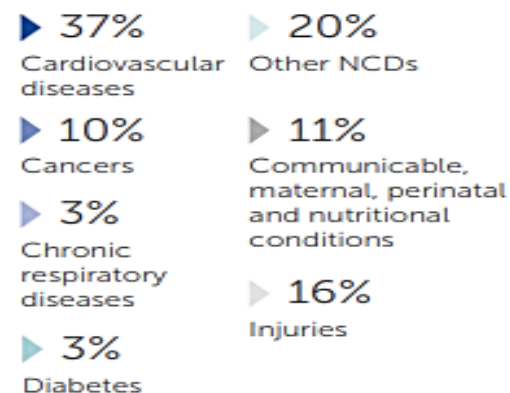
SAUDI ARABIA

RISK OF PREMATURE DEATH DUE TO NCDs (%)*



2016 TOTAL POPULATION: 32 276 000
2016 TOTAL DEATHS: 114 000

PROPORTIONAL MORTALITY*



- Al-Nozha MM, Arafah MR, Al-Mazrou YY, Al-Maatouq MA, Khan NB, Khalil MZ, Al-Khadra AH, Al-Marzouki K, Abdullah MA, Al-Harhi SS, Al-Shahid MS, Nouh MS, Al-Mobeireek A. Coronary artery disease in Saudi Arabia. Saudi Med J. 2004 Sep;25(9):1165-71. PMID: 15448760.
- WHO - Noncommunicable diseases Saudi Arabia 2018 country profile

Etiology

**Atherosclerosis
is the most
common cause**

Atherosclerosis Risk Factors

Modifiable Risk Factors

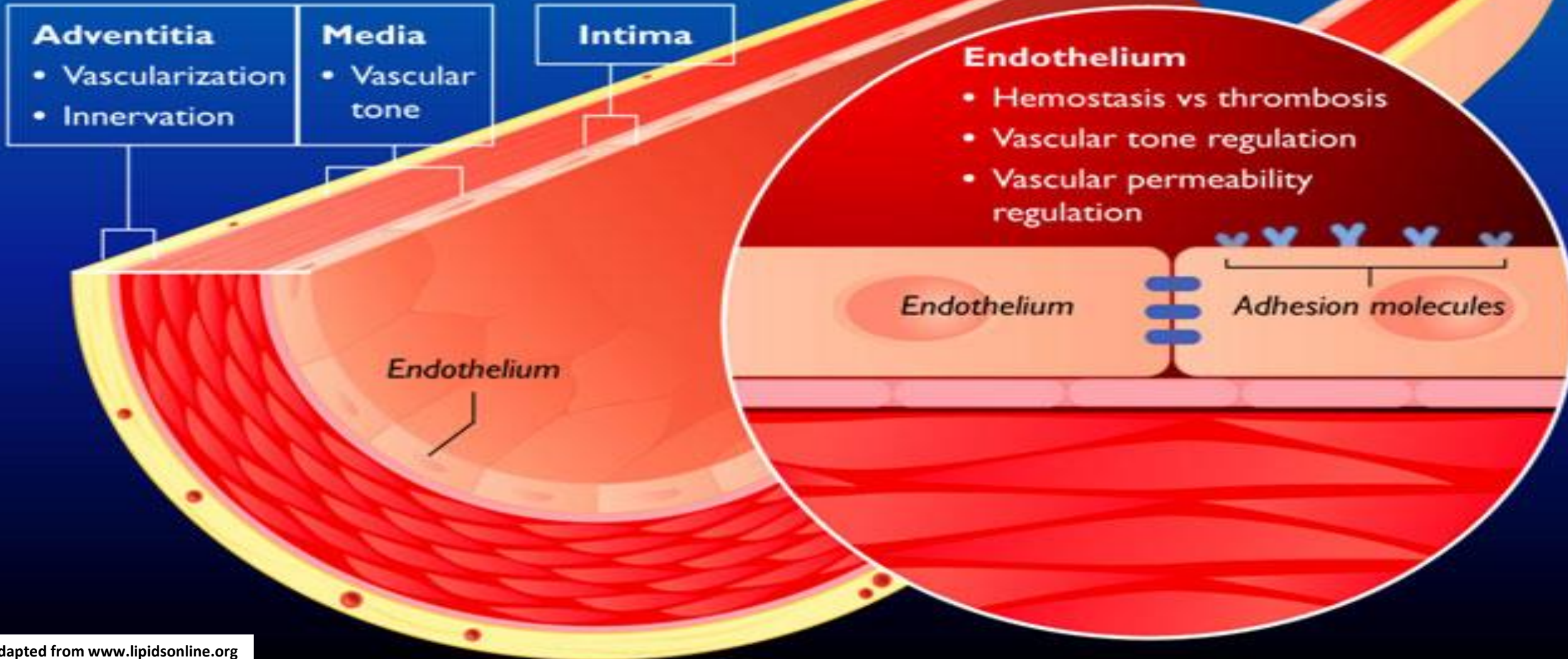
- Smoking
- DM
- HTN
- Dyslipidemia
- High homocysteine levels
- Obesity
- High fibrinogen levels
- Hyperphosphatemia
- Stress
- Increased alcohol consumption

Non-modifiable Risk Factors

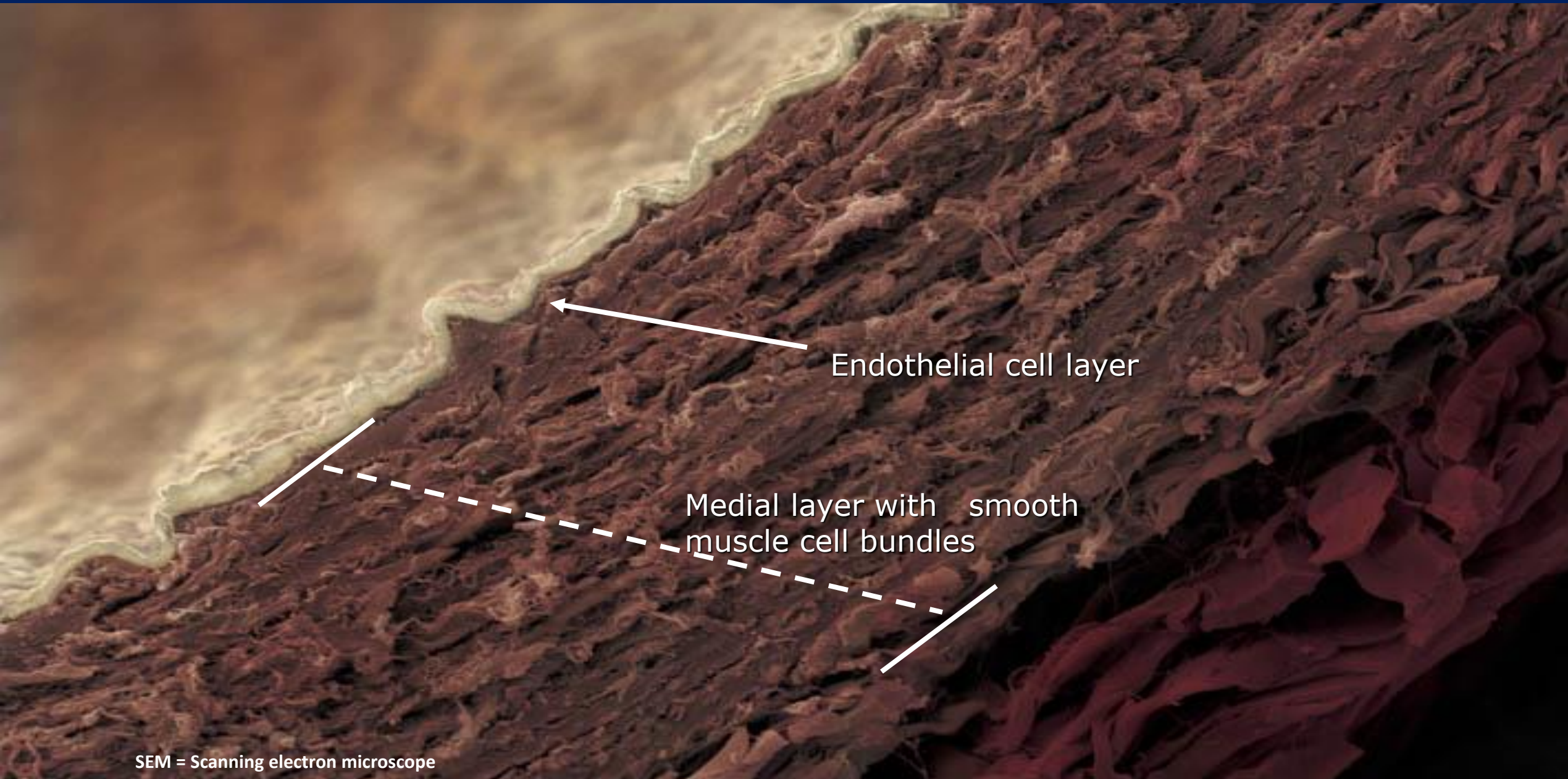
- Family history (cardiovascular events in:
 - **First-degree relative**
 - **Below the age of:**
 - **55 (♂)**
 - **65 (♀)**
- Age:
 - Males ≥ 45 years,
 - Females ≥ 55 years

The Arterial Wall

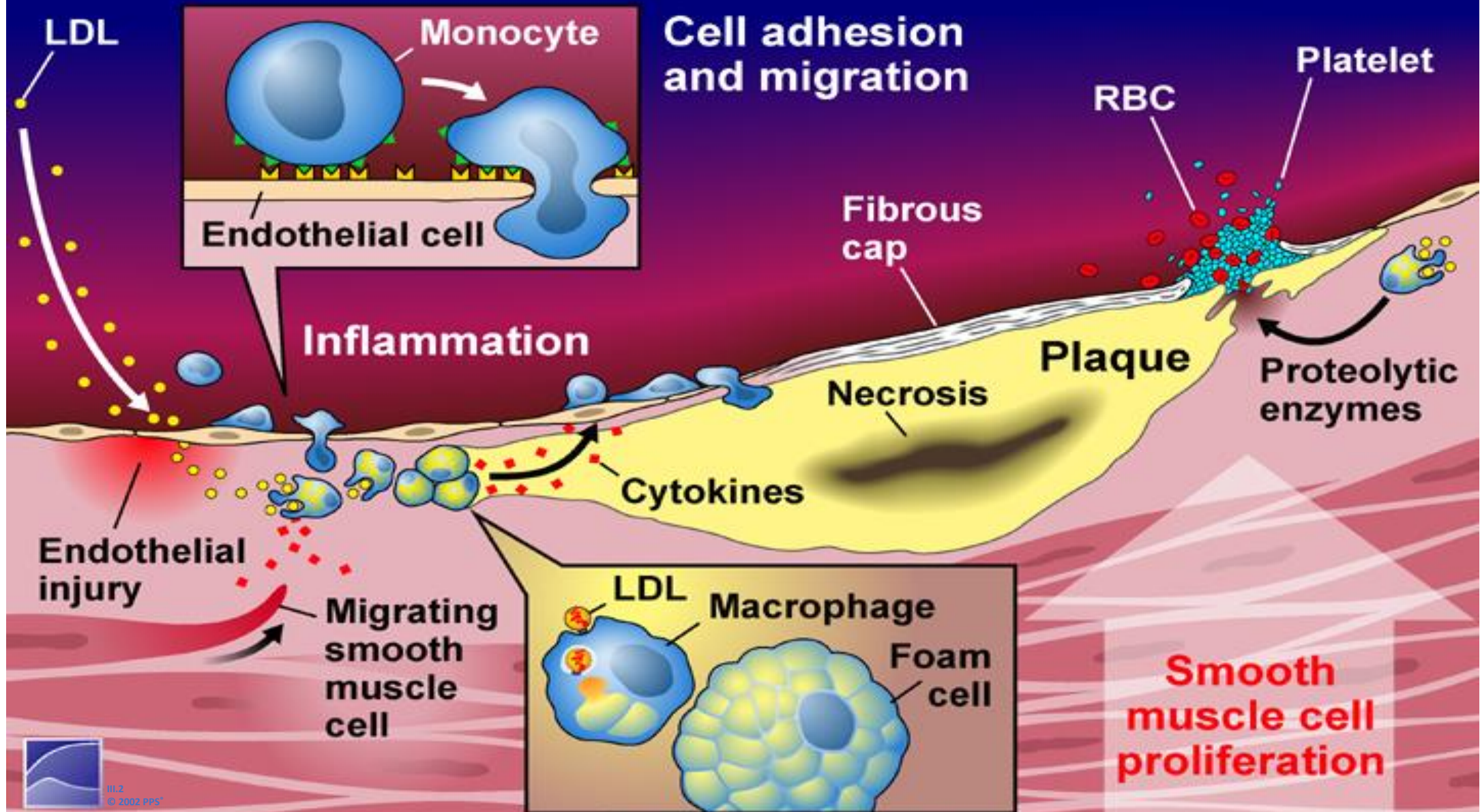
The artery has three structural components:

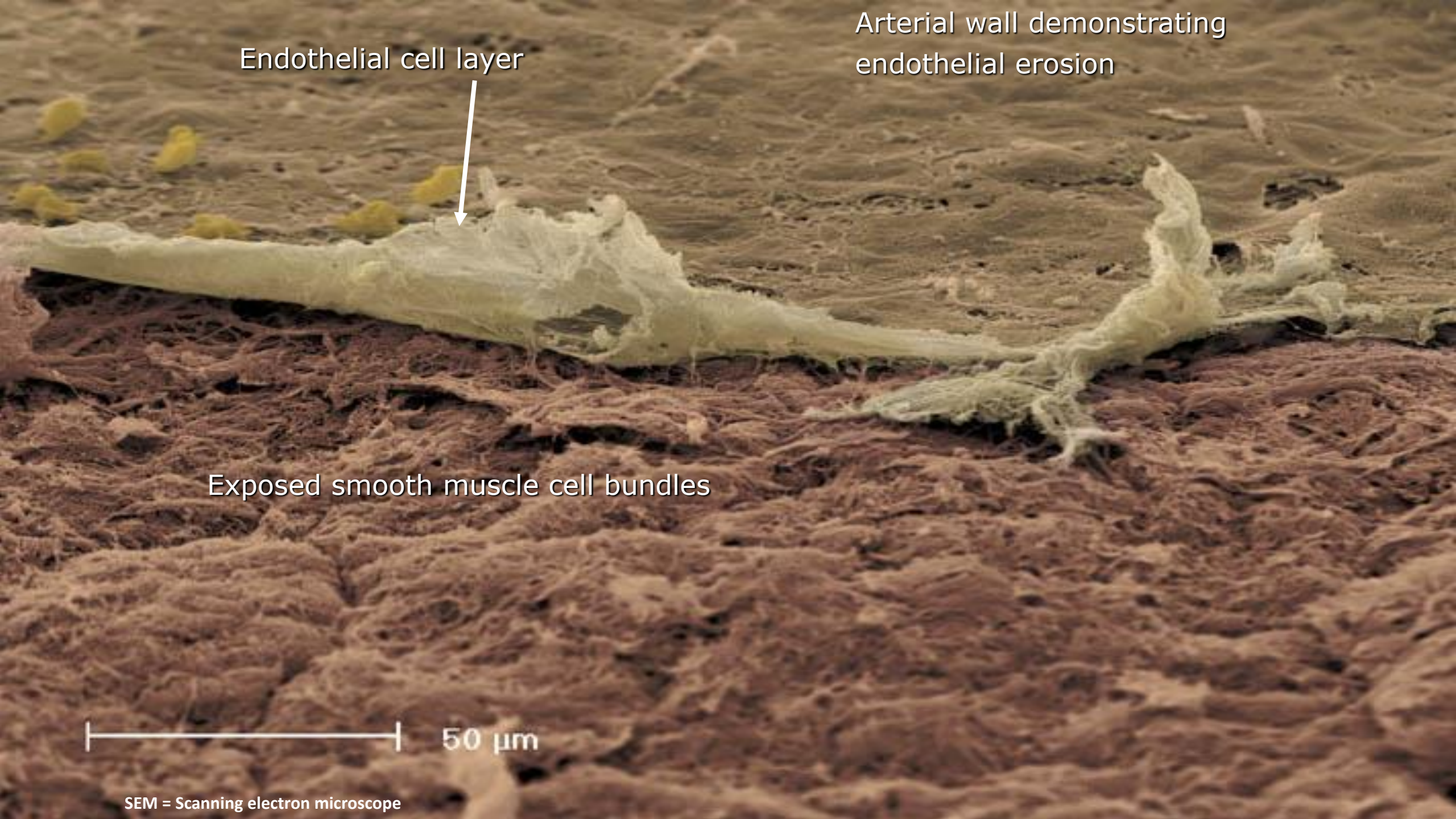


Cross-section through the wall of a healthy artery with intact endothelium, intima and smooth muscle bundles



Atherosclerosis: An Inflammatory Disease





Endothelial cell layer

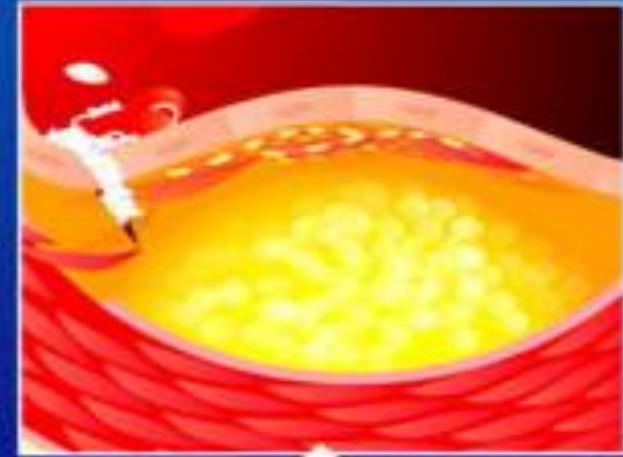
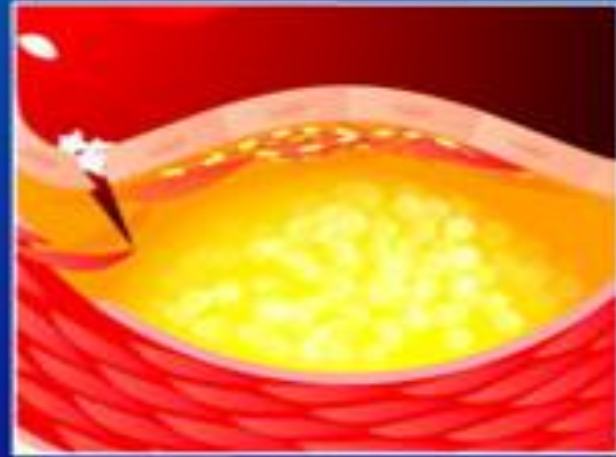
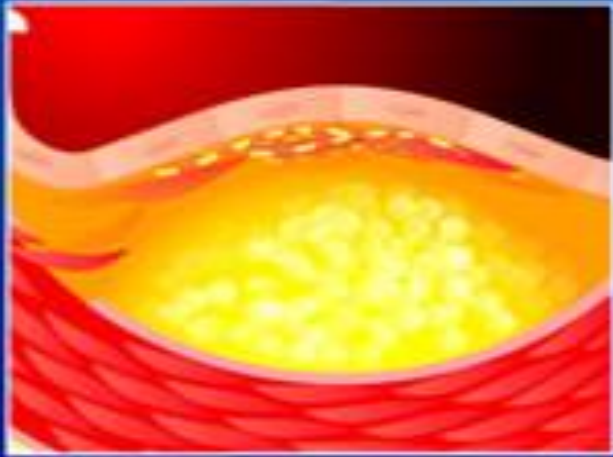
Arterial wall demonstrating
endothelial erosion

Exposed smooth muscle cell bundles

50 μm

SEM = Scanning electron microscope

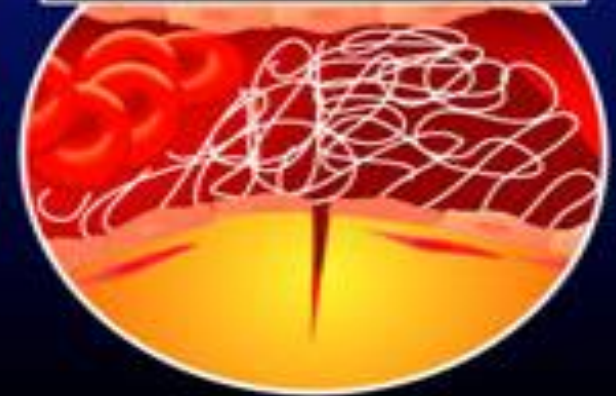
From Plaque to Thrombosis Plaque Rupture



**Partial occlusive
thrombosis**

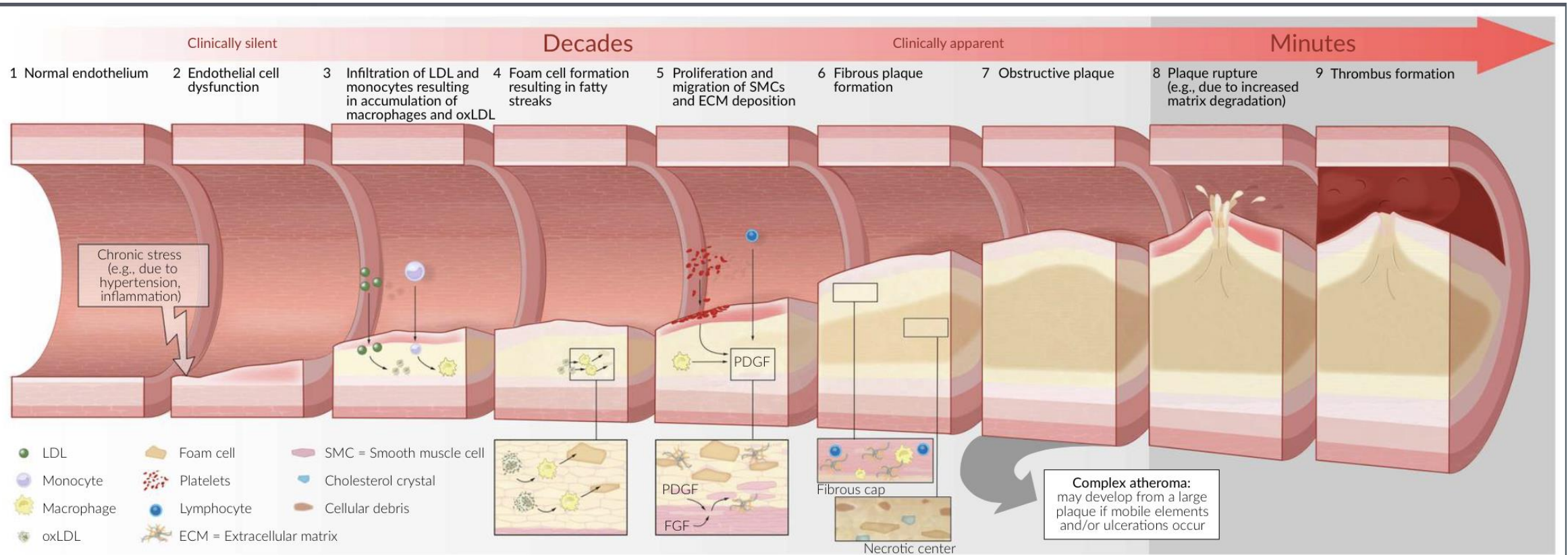


**Occlusive
thrombosis**



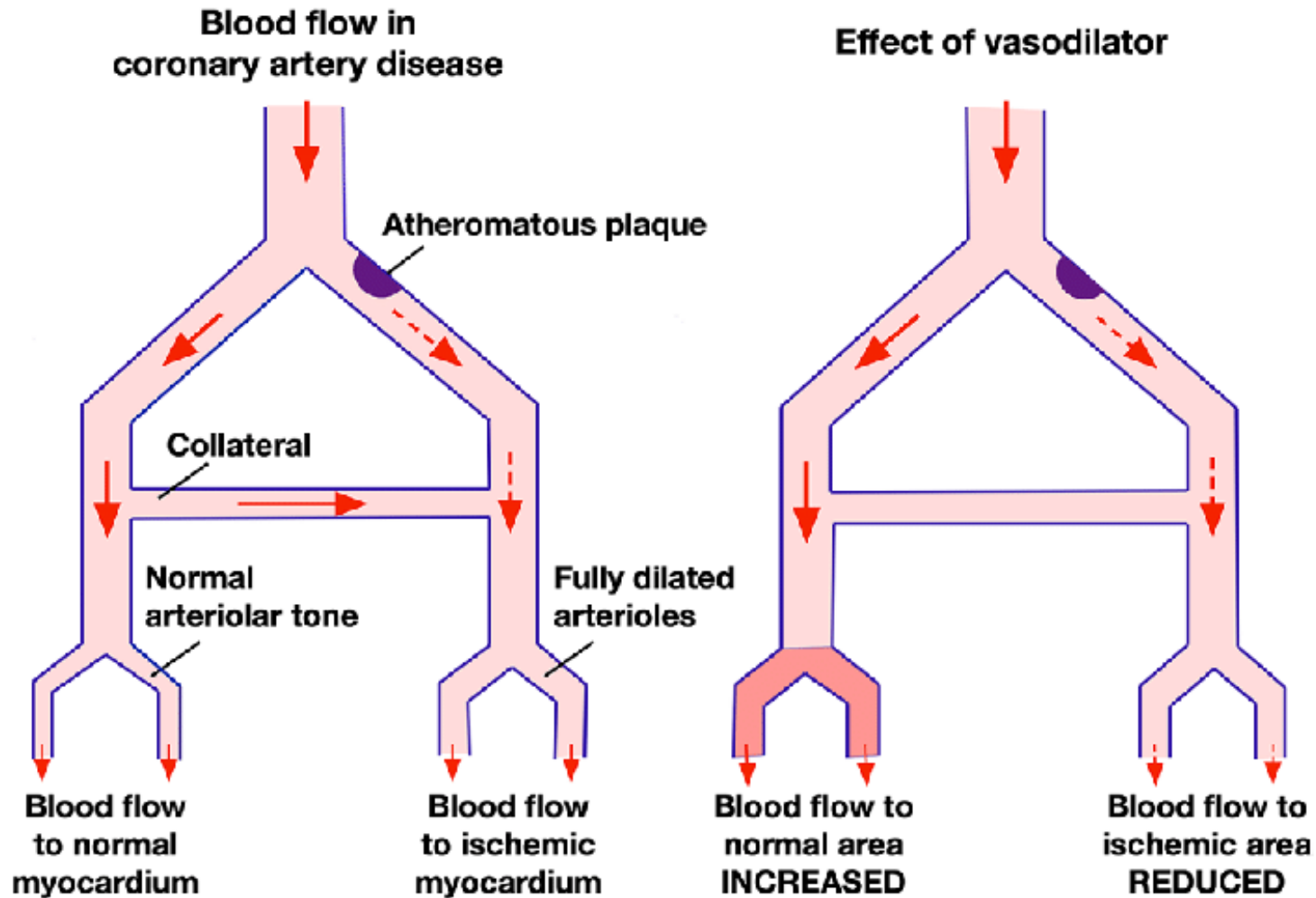
Pathophysiology

- Stable atherosclerotic plaque → vascular stenosis → increased resistance to blood flow in the coronary arteries → decreased myocardial blood flow → oxygen supply-demand mismatch → myocardial ischemia



Pathophysiology – Coronary Steal Syndrome

Coronary steal phenomenon



Clinical Features

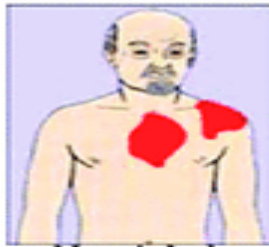
- Typically, **retrosternal chest pain** or pressure
 - May be absent, especially in geriatric, women, and diabetic patients
- Dyspnea
- Dizziness, palpitations
- Restlessness, anxiety
- Autonomic symptoms (e.g., diaphoresis, nausea, vomiting, syncope)



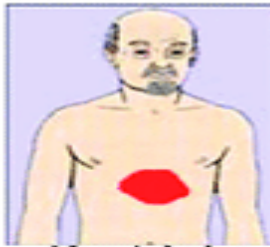
Upper chest pain



Chest pain radiates to jaw and neck



Chest Pain radiates at left arm



Chest pain radiates to epigastrium



Epigastrium pain radiates to arms and neck



Feeling like suffocating in the neck



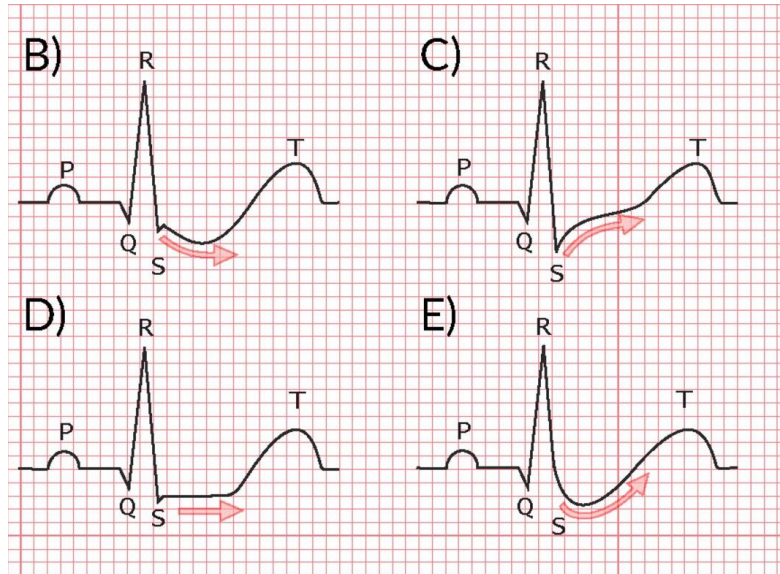
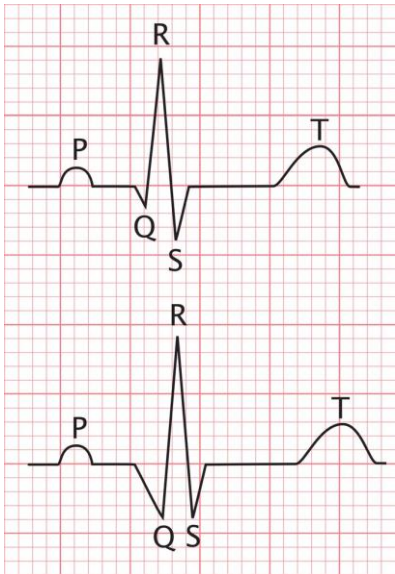
Left arm and both hands pain



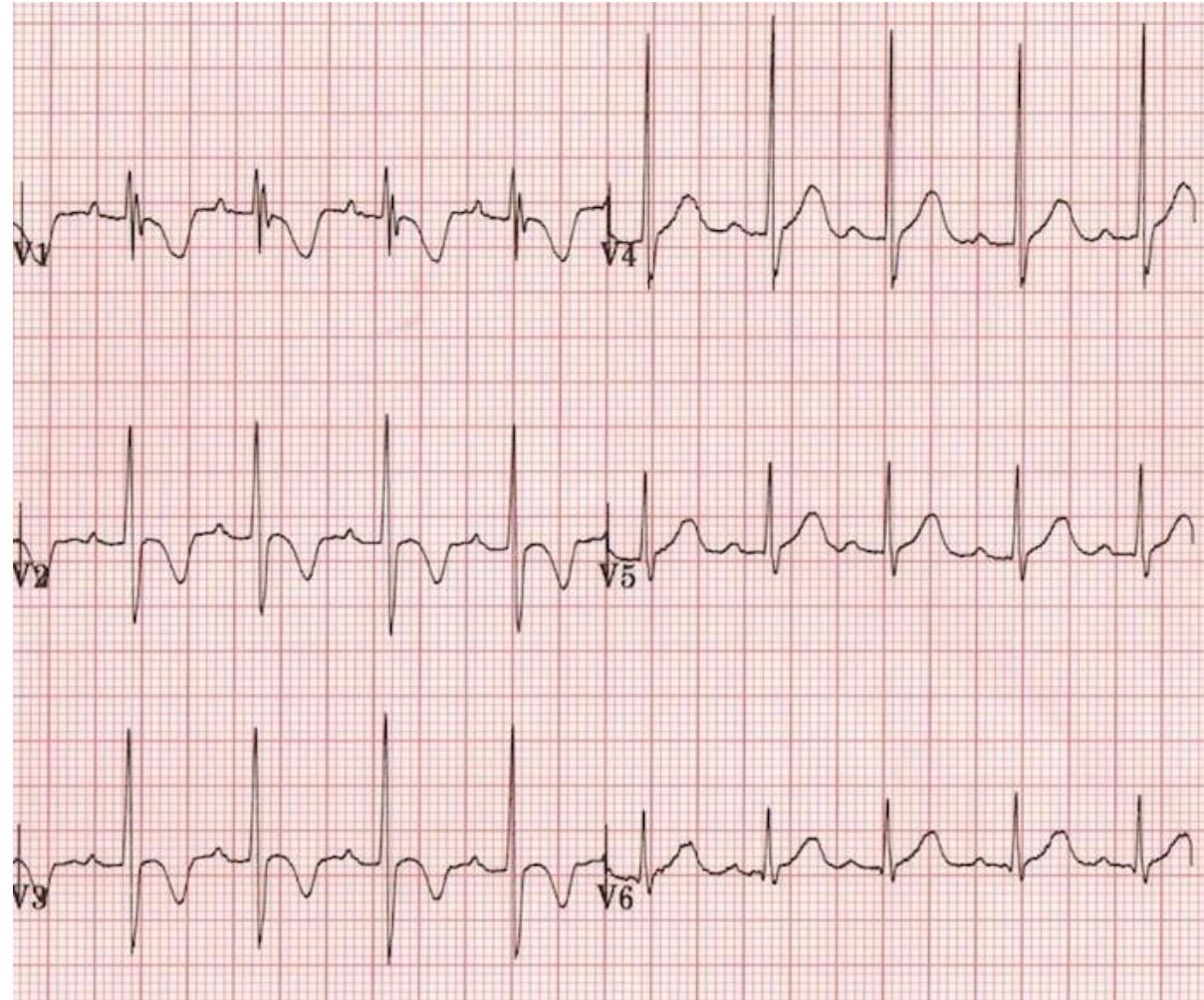
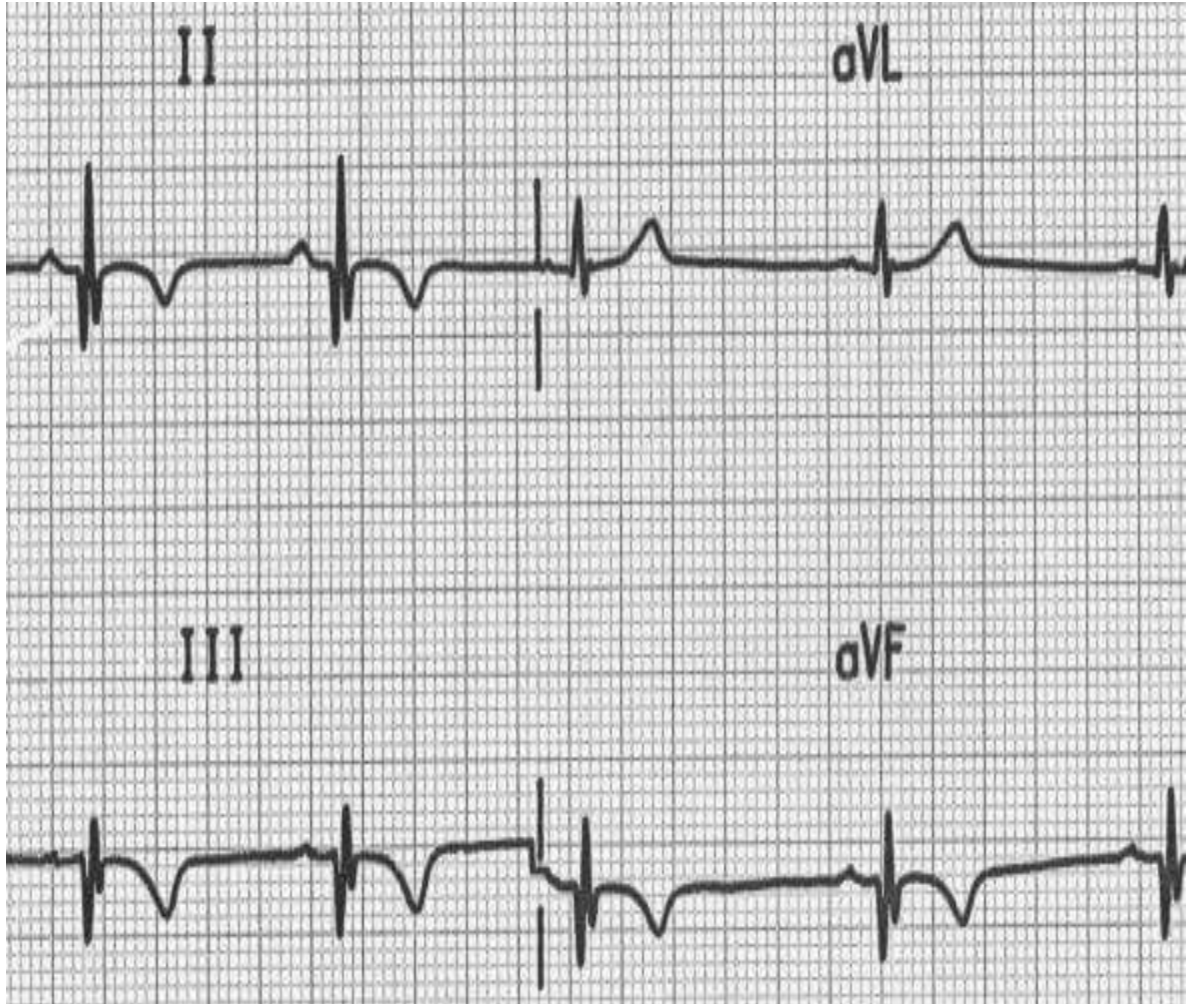
Pain at the middle of the back

Stable Angina – Diagnostics

- ECG is the best initial test
- Usually normal in patients with stable CAD
- Q waves (prior MI)
- Nonspecific T-wave changes (inversion, or flattening)



Stable Angina – Diagnostics



Stable Angina – Diagnostics

Pretest Probability of CAD

Estimated PTP	Clinical presentation	Recommendation
Low (<10%)	- Asymptomatic people of all ages - Atypical chest pain in women age < 50	Further testing is often unnecessary
Intermediate (20-80%)	- Atypical angina in men of all ages - Atypical angina in women age ≥ 50 - Typical angina in women age 30-50	Cardiac exercise stress test indicated
High (>90%)	- Typical angina in men age ≥ 40 - Typical angina in women age ≥ 60	Cardiac exercise stress test indicated Vs. Proceed with cardiac catheterization

Stable Angina – Diagnostics

Cardiac stress testing:

Detect stress induced ischemia

- **Indications:** CAD and non-coronary artery disease
- **How it works:** Type of stress and stress diagnostics
- **The goal** is to detect evidence of stress-induced ischemia by targeting 85% of the maximum heart rate ($220 - \text{age in years}$)

Types of Stress



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Exercise

Everyone who can exercise should!
They must be able to raise their heartrate to 85% of predicted value.

Predicted HR = 220 bpm - age

Pharmacologics

Vasodilators

Ex. Adenosine, Dipyridamole, Regadenoson

↑ cAMP production to cause vascular smooth muscle relaxation & coronary vasodilation

Contraindications:
Bradycardia & hypotension. Avoid in pts w/ 2nd or 3rd heartblock w/o pacemaker
Bronchoconstriction. Avoid in pts with severe COPD and asthma

Inotropes

Ex. Dobutamine
Beta 1 and 2 agonist

Contraindications:
Pts with ventricular arrhythmias



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Stress Diagnostics

EKG (\$)

Sensitivity / Specificity	
EKG + Exercise	61-68% / 70-77%
PROS	
No radiation Easy to obtain	
CONS	
Low specificity and sensitivity Cannot localize ischemia EKG alone not good enough for: V-paced, WPW, ST Depression, Digoxin, LBBB Volkswagen Drivers Don't Litter	

ECHO (\$\$)

Sensitivity / Specificity	
ECHO + EKG + Exercise	70-85% / 77-89%
PROS	
No radiation High specificity and sensitivity Can detect valvular pathologies, LVOT obstruction, & pulmonary HTN	
CONS	
Img quality is technician & patient dependent	

Nuclear SPECT (\$\$\$)

Sensitivity / Specificity	
Nuclear SPECT + Exercise	82-88% / 70-88%
Nuclear SPECT + Pharm	88-91% / 75-90%
PROS	
High specificity and sensitivity	
CONS	
Img quality is technician & reader dependent Balanced ischemia	

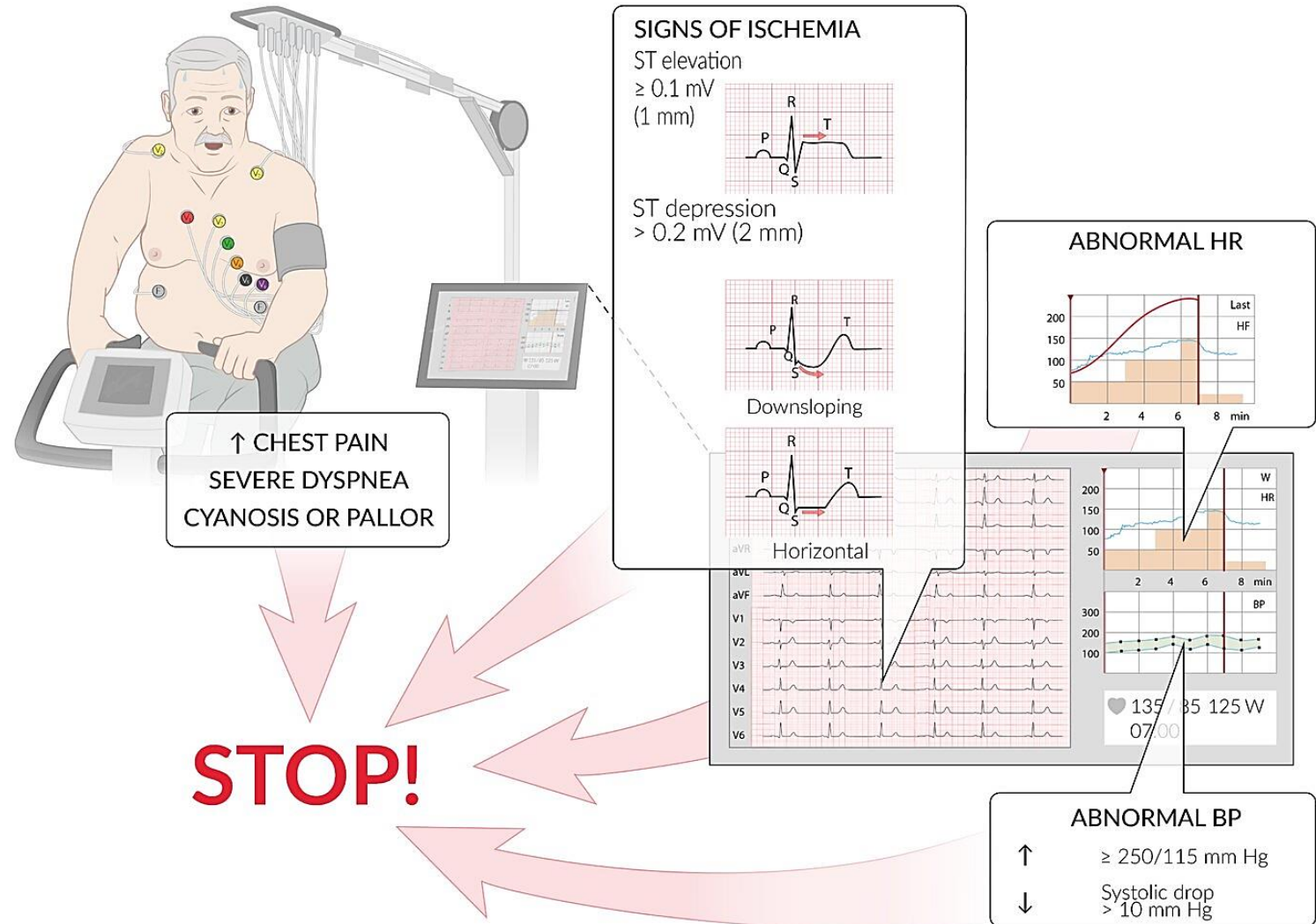
PET CT/MRI (\$\$\$\$)

Sensitivity / Specificity	
PET CT/MRI + Pharm	92-93% / 83-85%
PROS	
High specificity and sensitivity	
CONS	
Less easily available Costly	

Stable Angina – Diagnostics

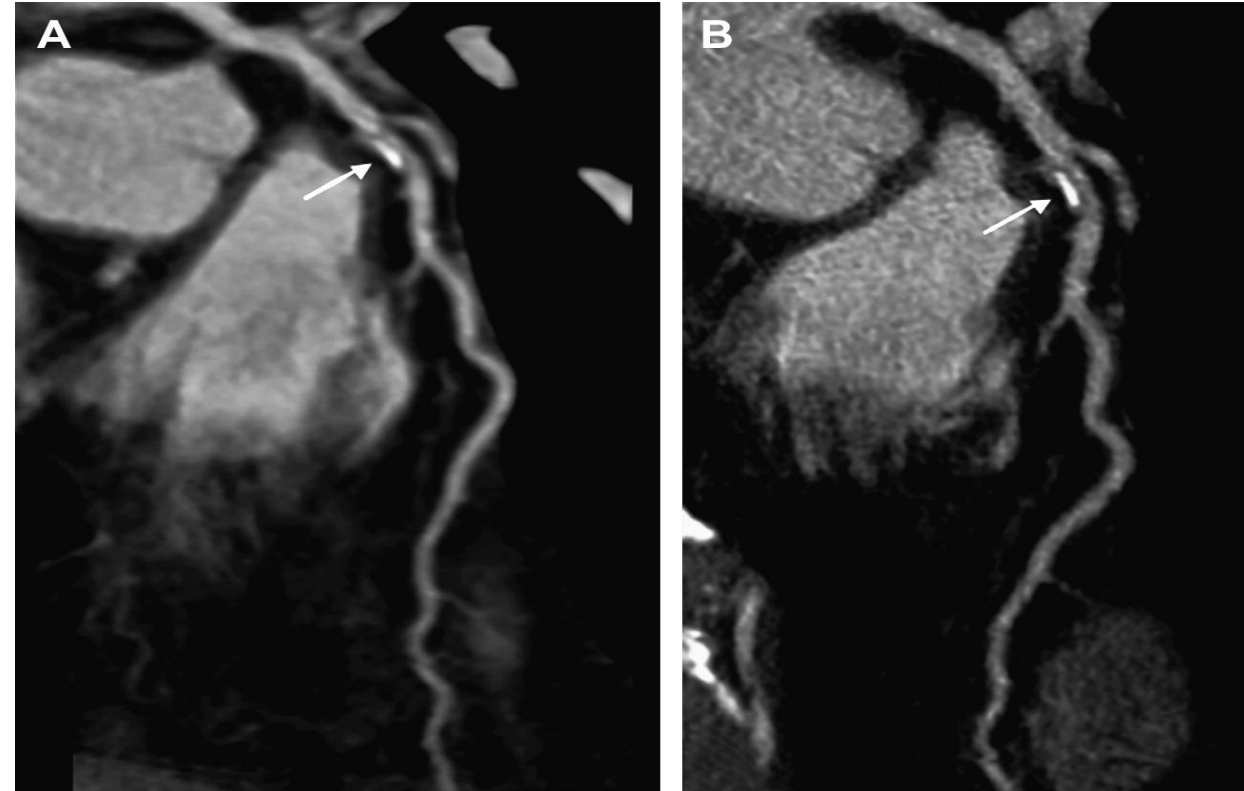
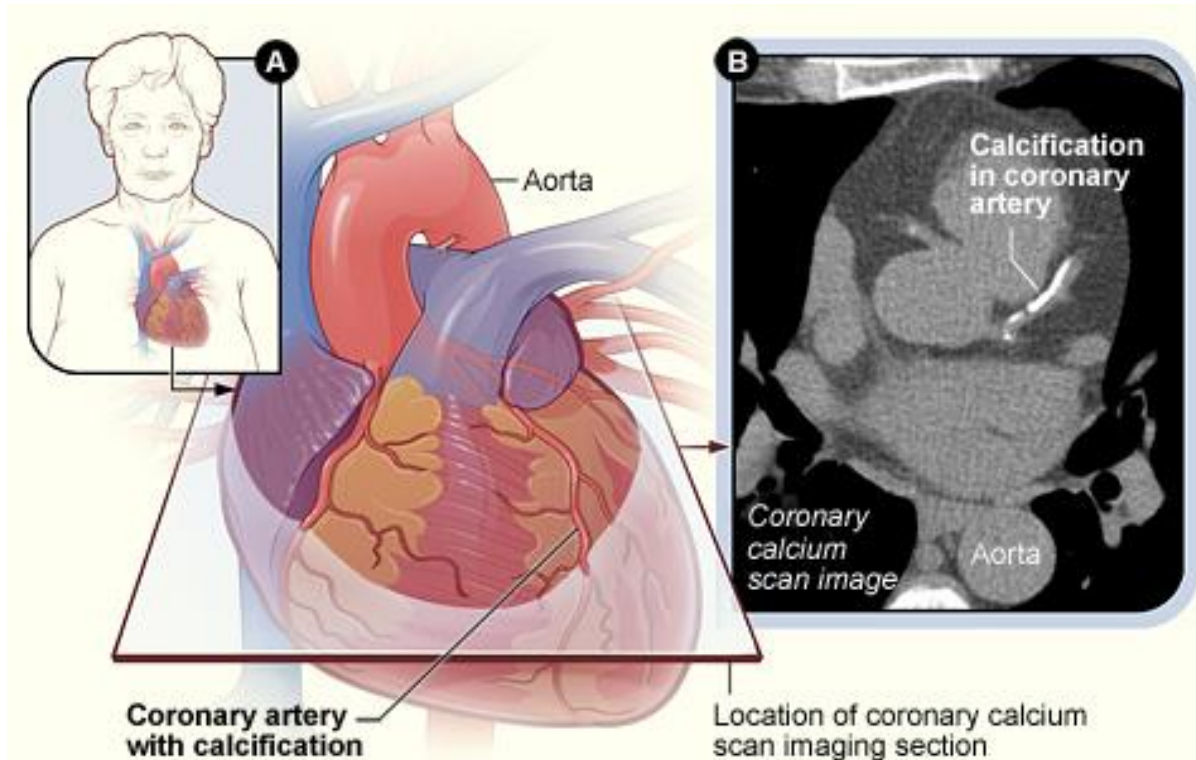
Test is terminated by:

- Reaching a diagnostic endpoint
- Reaching target HR
(no diagnostic endpoint)
- Significant cardiac arrhythmia
- Patient inability to complete



Stable Angina – Diagnostics

- Coronary CT angiography
- Coronary artery calcium (CAC) scoring

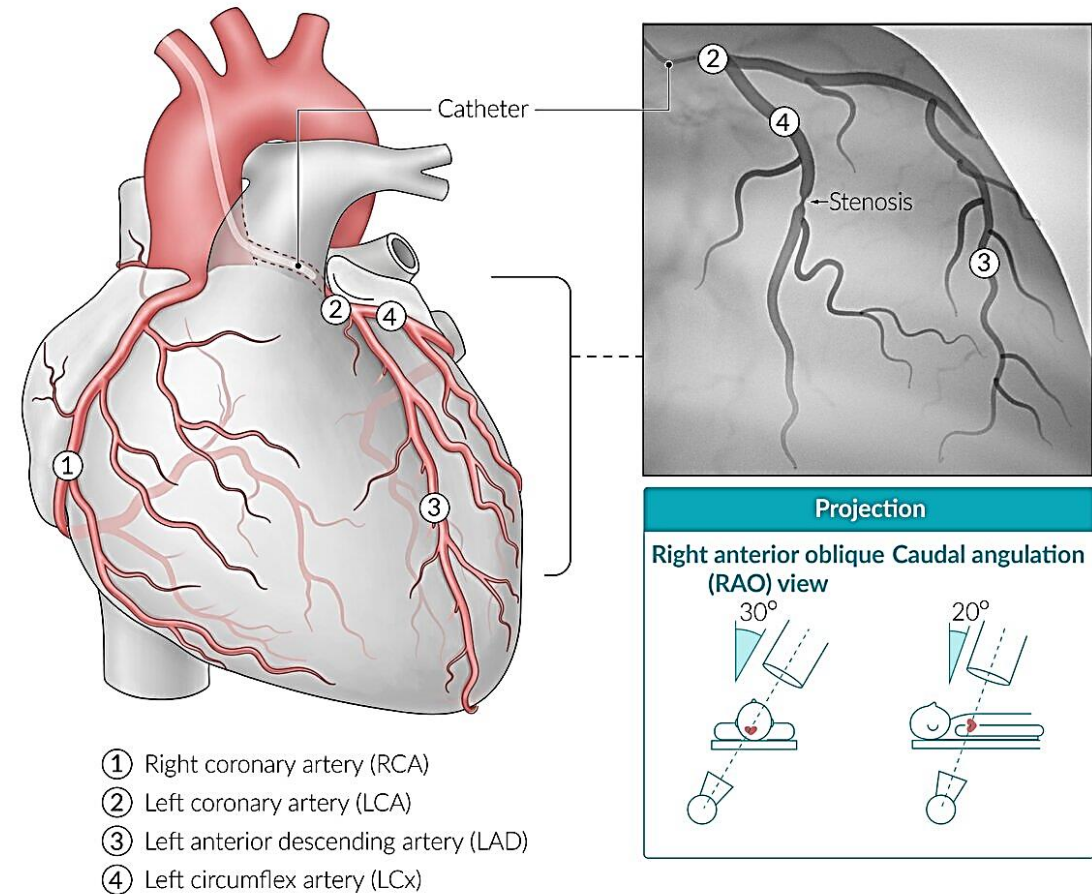


Stable Angina – Diagnostics

- Cardiac catheterization is the gold standard

Indications:

- Acute MI
- A positive stress test
- Noninvasive tests are non-diagnostic/contraindicated
- Angina that occurs despite medical therapy



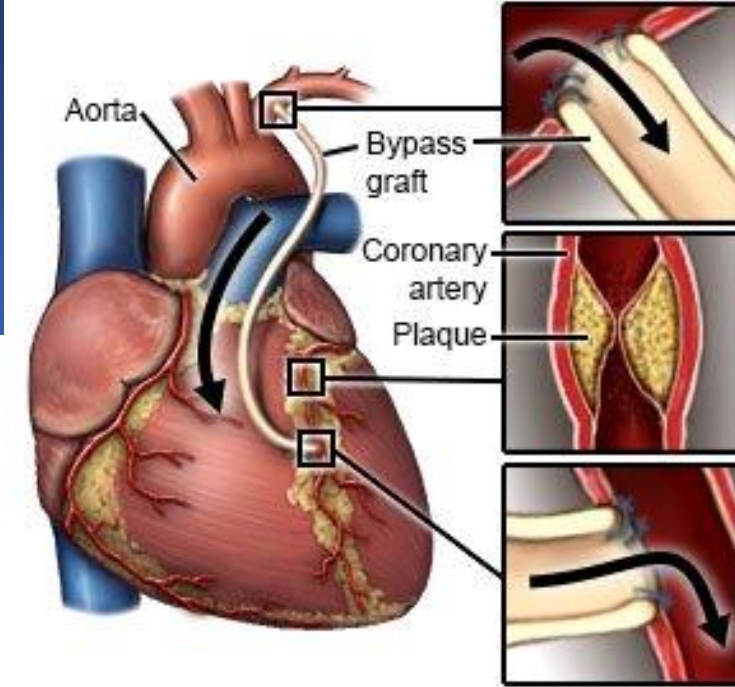
Stable Angina – Management

All patients will receive pharmacotherapy for CAD

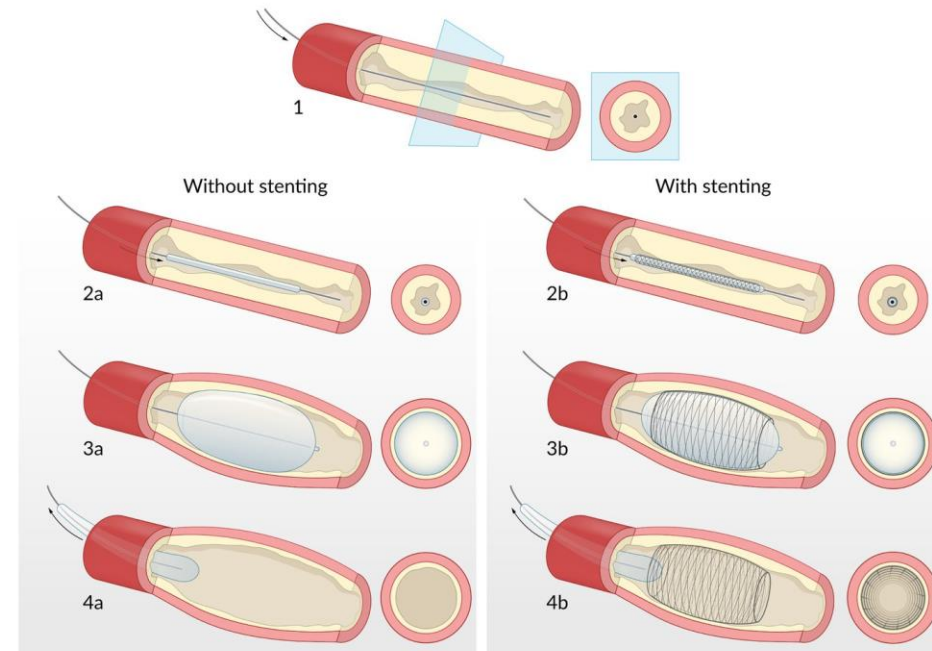
- Management of comorbidities
- Secondary prevention of CAD
- Antianginal medication

Select patients will receive revascularization

- Percutaneous coronary intervention (PCI)
- Coronary artery bypass grafting (CABG)



Coronary Artery Bypass Surgery



Stable Angina – Management

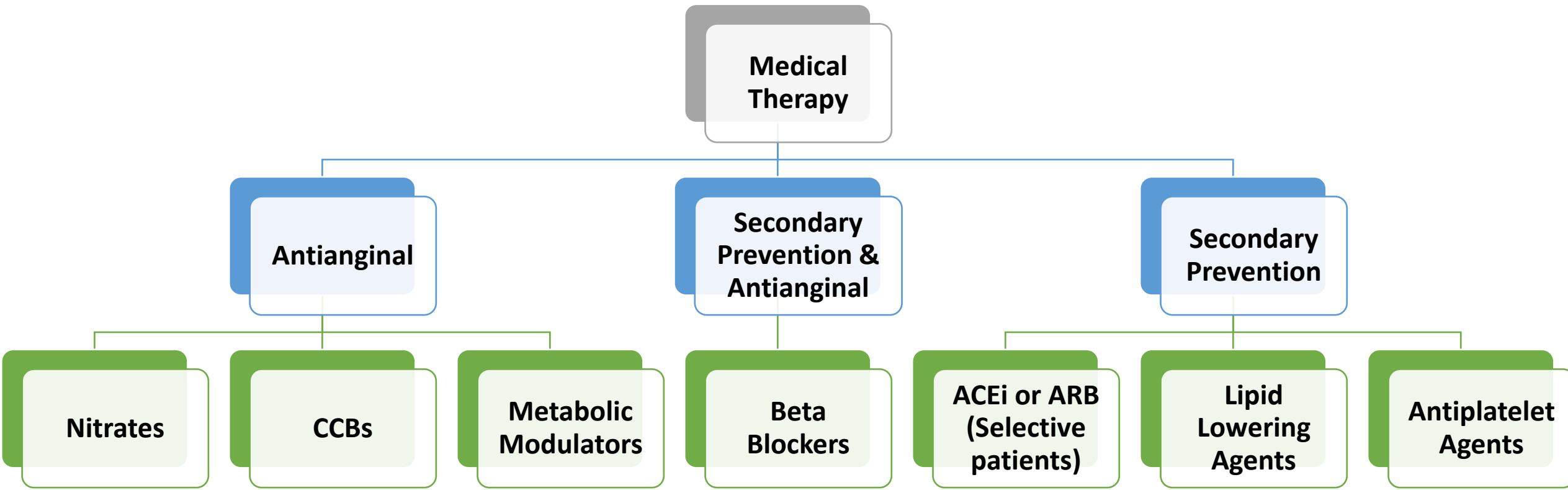
All patients should be counseled on Lifestyle modifications

- Smoking cessation
- Increased physical activity

All patients receive treatment of comorbidities

- Hypertension (Target BP < 130/80 mm Hg)
- Diabetes mellitus (Individualized glycemic goals but ideally HbA1c < 7%)
- Hyperlipidemia (goal LDL < 70 mg/dL)

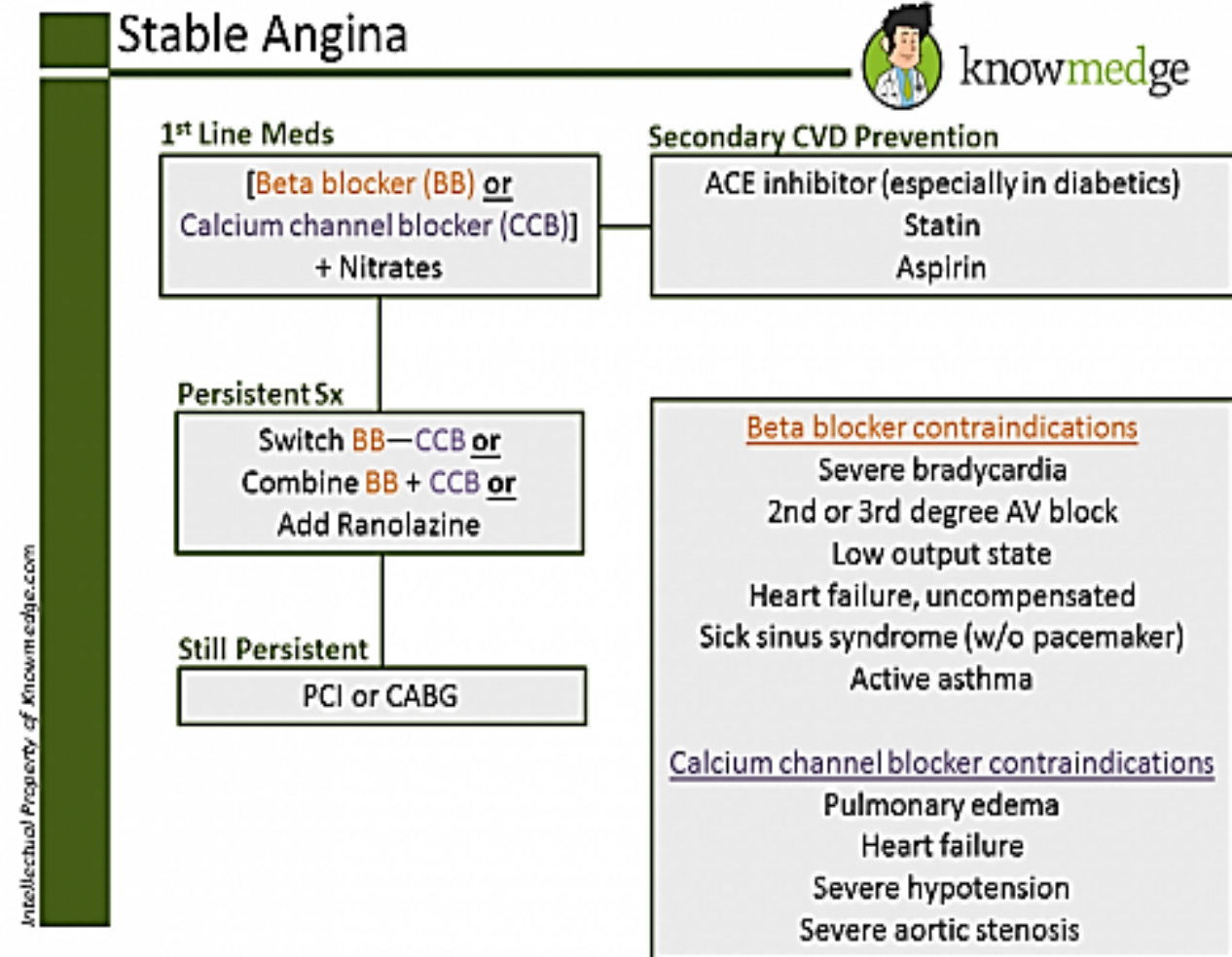
Stable Angina – Management



Stable Angina – Management

Secondary prevention medications:

- Aspirin
- Clopidogrel (A P2Y12 receptor antagonist) in patients with contraindications to aspirin
- ACEi and ARBs (HTN, DM, CKD, LVEF < 40%)
- High-intensity statin therapy
- PCSK9 inhibitors if LDL not at goal with statin



Stable Angina – Management

Revascularization therapy:

- Controversial for patients with stable CAD
- COURAGE trial suggest no improvement in mortality and MI (when compared to medical therapy alone)
- Decided by cardiologist



Optimal Medical Therapy with or without PCI for Stable Coronary Disease

William E. Boden, M.D., Robert A. O'Rourke, M.D., Koon K. Teo, M.B., B.Ch., Ph.D., Pamela M. Hartigan, Ph.D., David J. Maron, M.D., William J. Kostuk, M.D., Merrill Knudtson, M.D., Marcin Dada, M.D., Paul Casperson, Ph.D., Crystal L. Harris, Pharm.D., Bernard R. Chaitman, M.D., Leslee Shaw, Ph.D., Gilbert Gosselin, M.D., Shah Nawaz, M.D., Lawrence M. Tittle, M.D., Gerald Gau, M.D., Alvin S. Blaustein, M.D., David C. Booth, M.D., Eric R. Bates, M.D., John A. Spertus, M.D., M.P.H., Daniel S. Berman, M.D., G.B. John Mancini, M.D., and William S. Weintraub, M.D., for the COURAGE Trial Research Group*

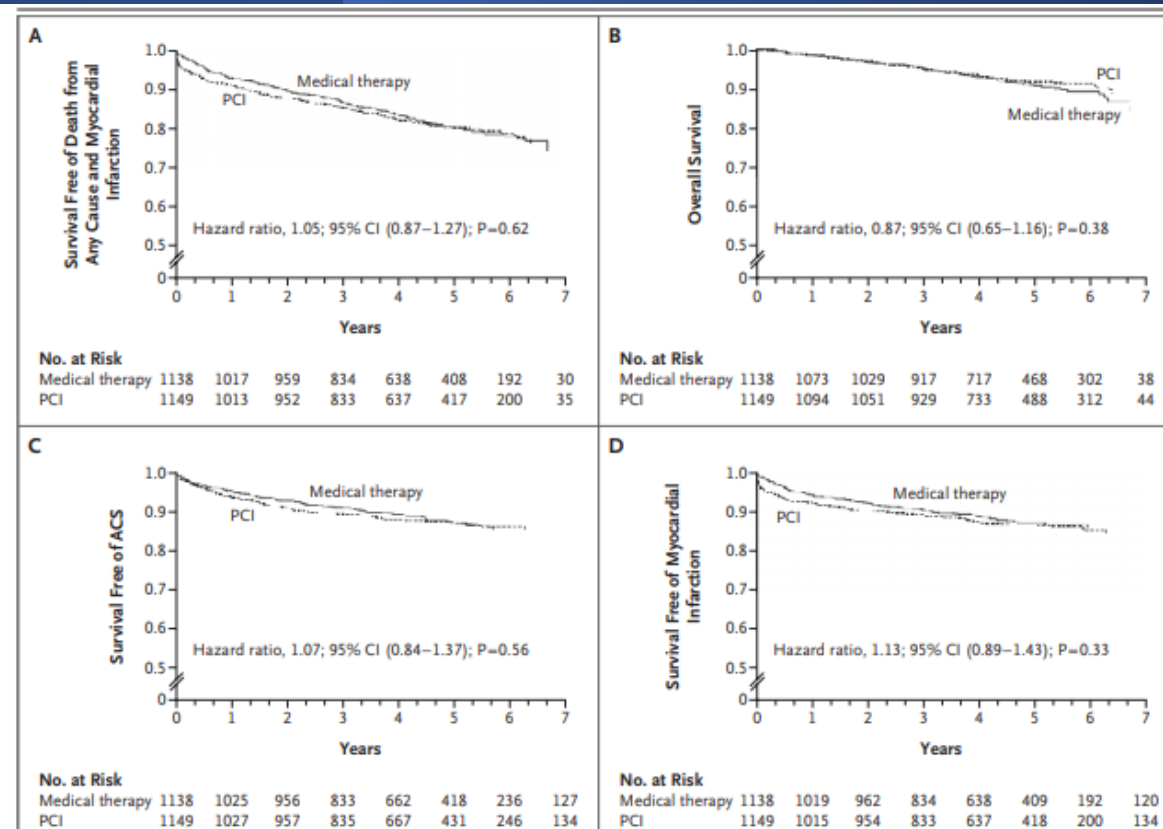
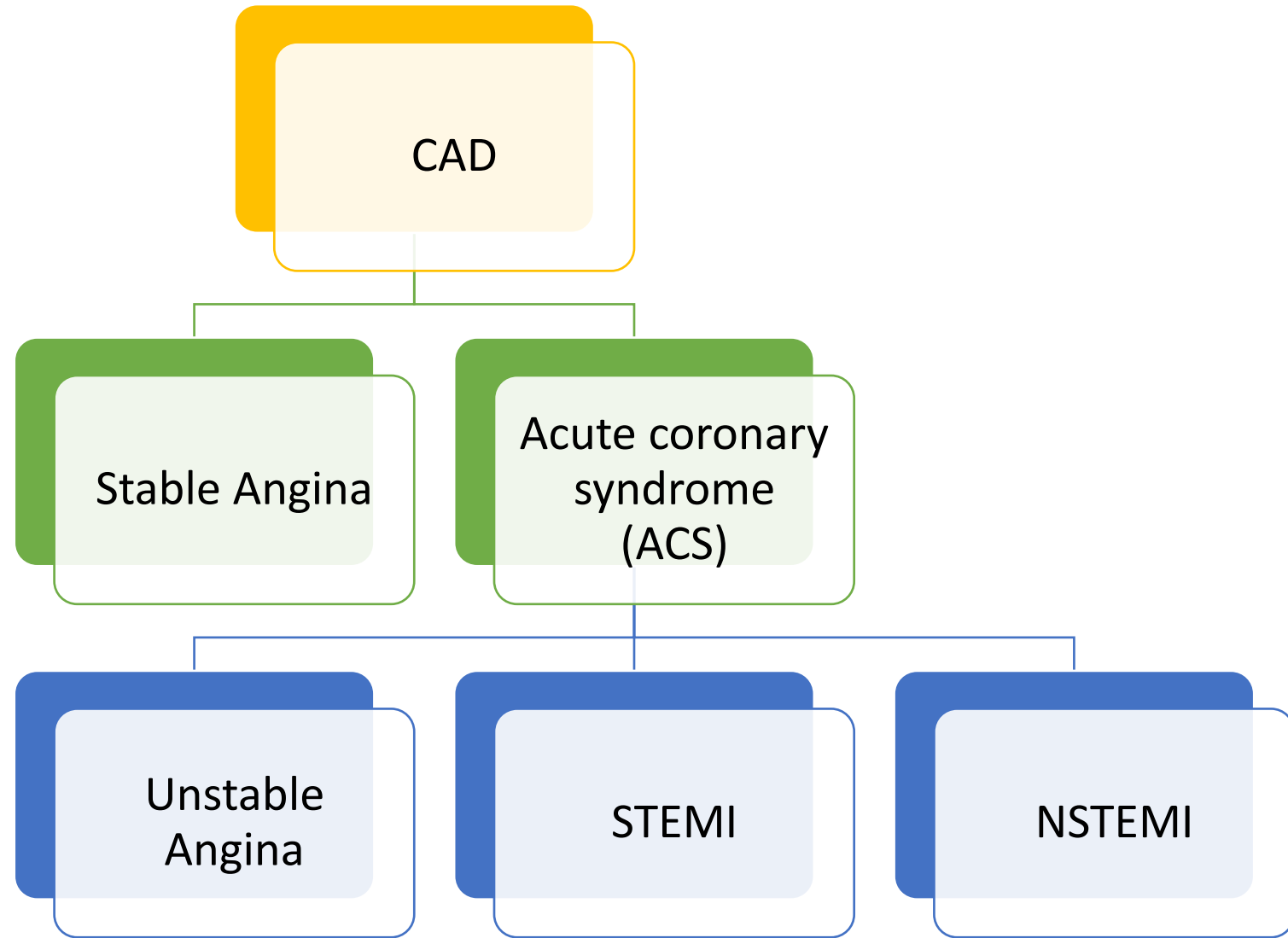


Figure 2. Kaplan-Meier Survival Curves.

In Panel A, the estimated 4.6-year rate of the composite primary outcome of death from any cause and nonfatal myocardial infarction was 19.0% in the PCI group and 18.5% in the medical-therapy group. In Panel B, the estimated 4.6-year rate of death from any cause was 7.6% in the PCI group and 8.3% in the medical-therapy group. In Panel C, the estimated 4.6-year rate of hospitalization for acute coronary syndrome (ACS) was 12.4% in the PCI group and 11.8% in the medical-therapy group. In Panel D, the estimated 4.6-year rate of acute myocardial infarction was 13.2% in the PCI group and 12.3% in the medical-therapy group.

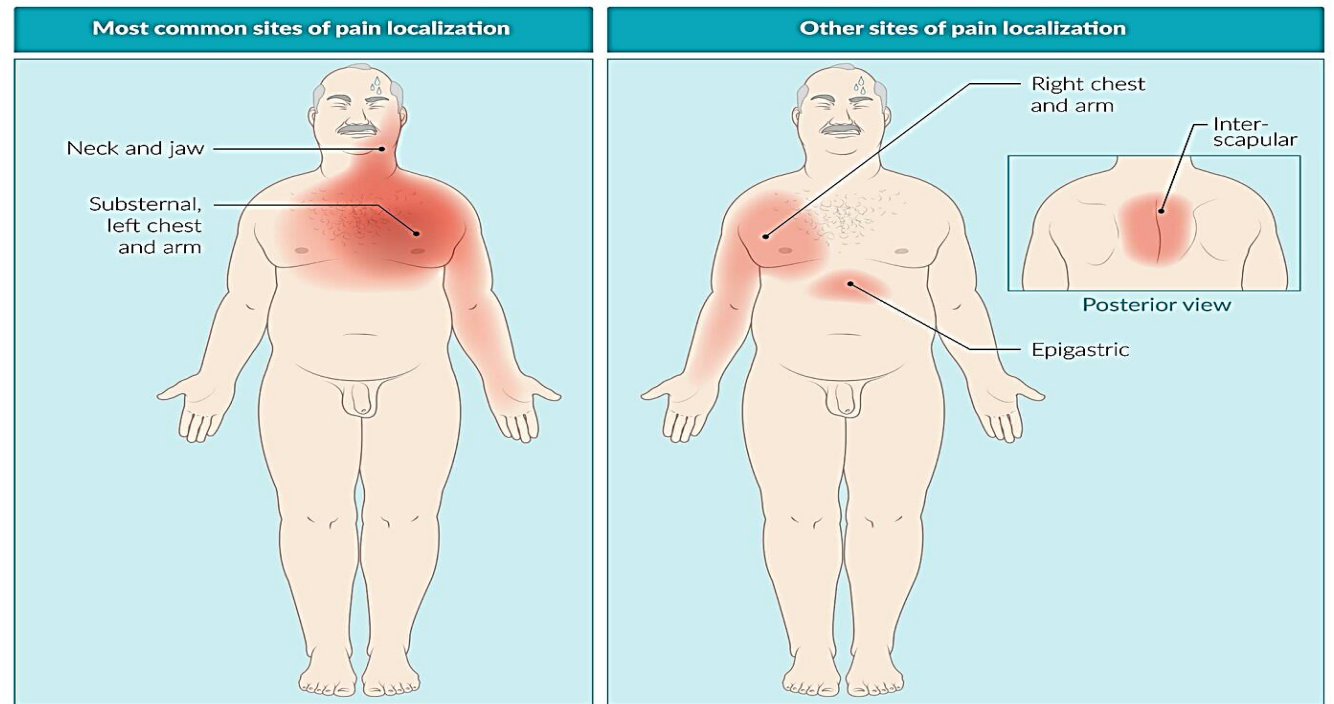
ACS

- **Unstable angina:** absence of detectable myocardial injury biomarkers
- **NSTEMI:** positive myocardial injury biomarkers **without** ST-elevations on ECG
- **STEMI:** positive myocardial injury biomarkers **with** ST-elevations on ECG



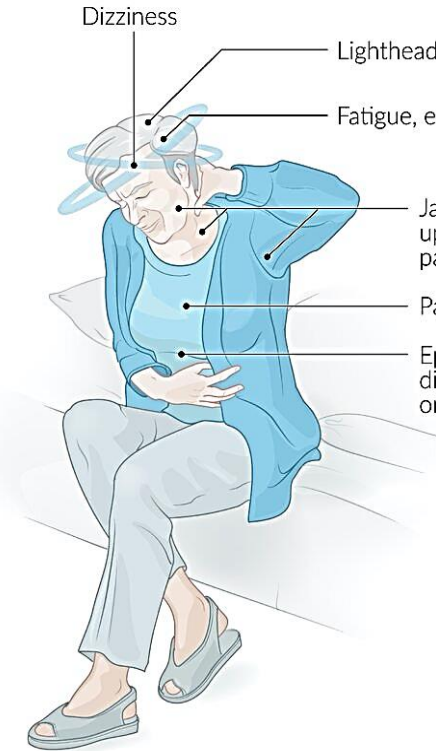
ACS – Clinical Features

- Chest pain (dull, squeezing pressure, acute, retrosternal)
- Precipitated by stress
- Dyspnea
- Pallor
- Nausea
- Vomiting
- Diaphoresis, anxiety
- Dizziness, lightheadedness, syncope



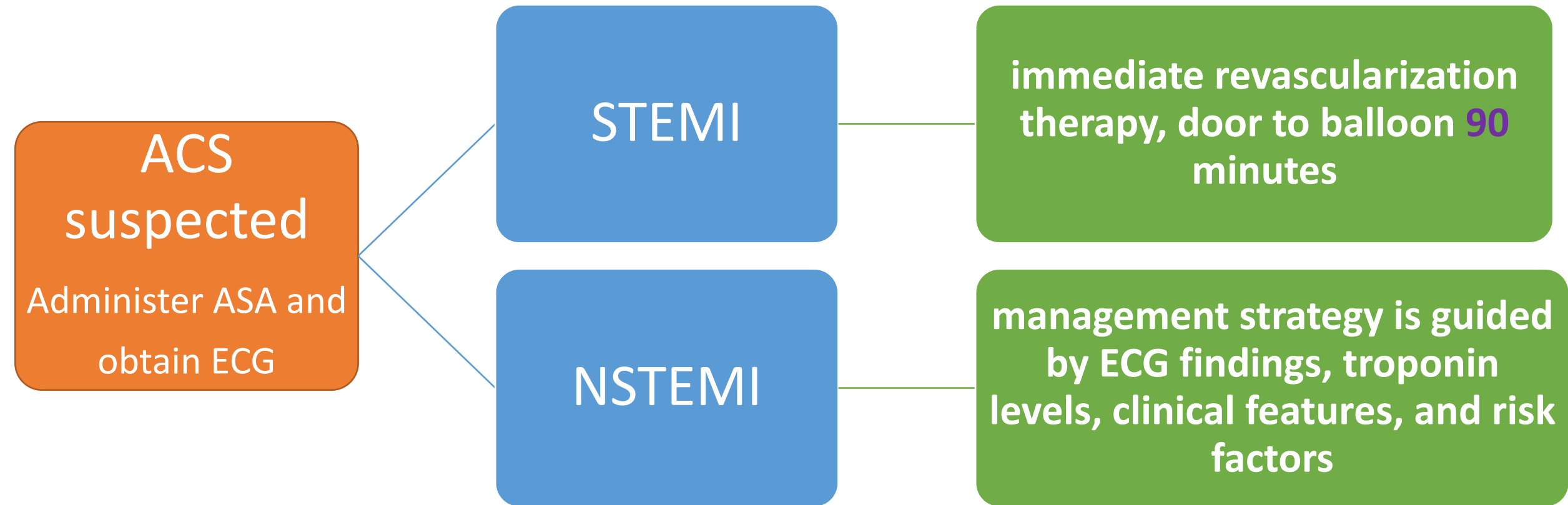
ACS – Clinical Features

- Atypical presentations in the elderly, diabetics, and women
 - “Silent MI” in diabetics due to polyneuropathy
 - Clinical triad of
 - Hypotension
 - Elevated JVP
 - Clear lung fields
- (Seen with right ventricular infarction)

Cardinal symptoms	Unspecific warning signs, esp. in women
 Diaphoresis, sense of impending doom, restlessness Nausea and vomiting Dyspnea Retrosternal pain radiating to the left side of the body (arm, shoulder, neck, and epigastrium)	 Dizziness Lightheadedness, fainting Fatigue, exhaustion Jaw, neck or upper back pain Palpitations Epigastric discomfort or pressure

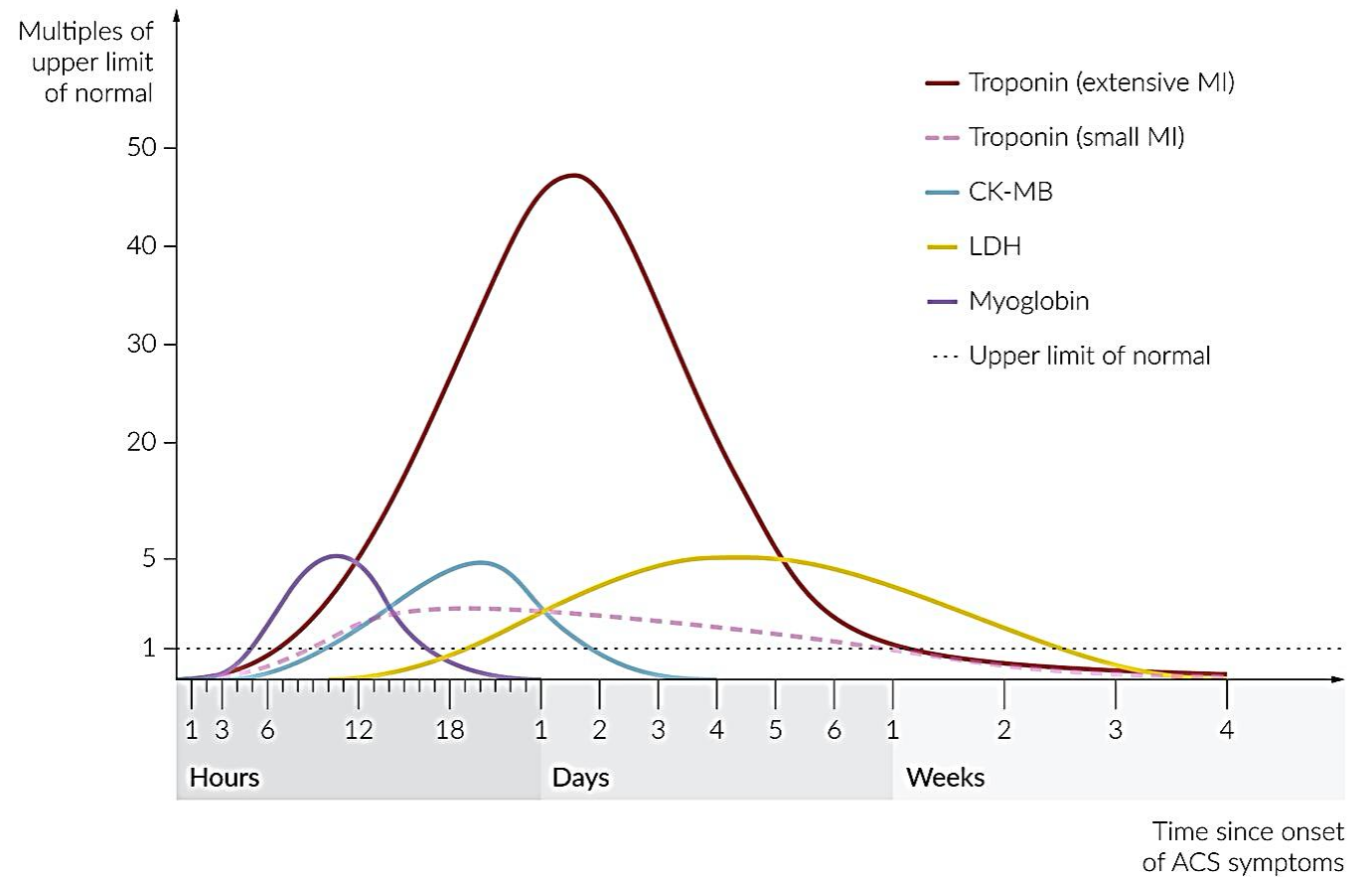
ACS – Diagnostics

- ECG should be performed **immediately** once ACS is suspected (**Best initial test**)



ACS – Diagnostics

- Troponin T/I is the most important cardiac marker
(**high sensitivity**)
- CK-MB can be useful in evaluating **reinfarction**
- Trend values till peaked
- Obtain if symptoms or ECG findings change



ACS – Diagnostics

Echocardiography is generally not necessary and should not delay reperfusion therapy

- **Indications:**

- Cardiogenic shock
- Infarct-like symptoms but inconclusive ECG findings
- Evaluation for complications of myocardial infarction

- **Findings:**

- Wall motion abnormalities
- Decreased LV function

ECG Changes in STEMI

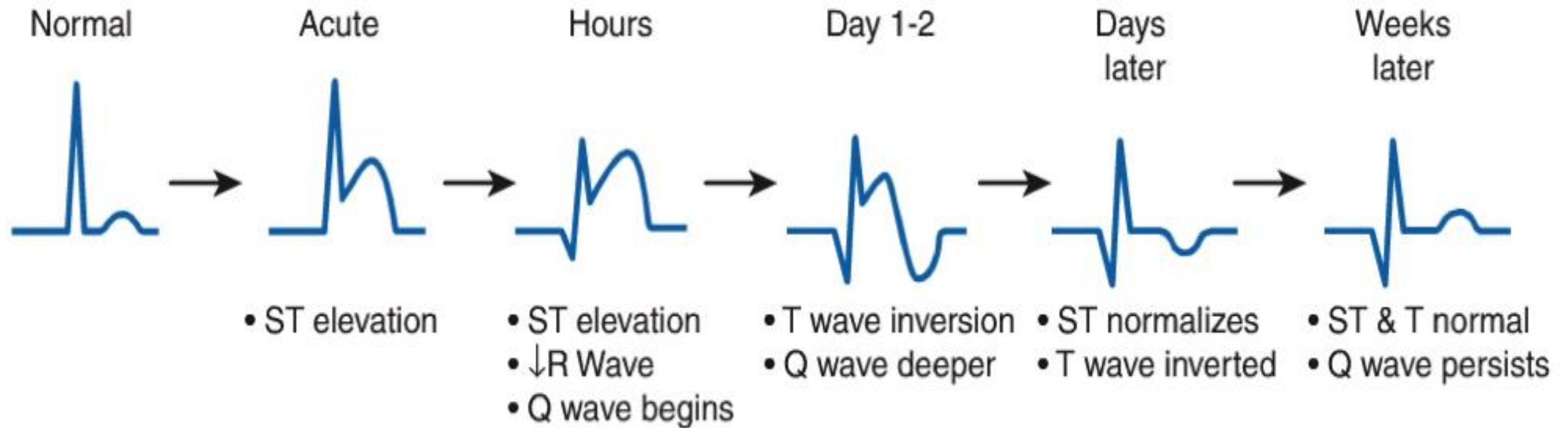
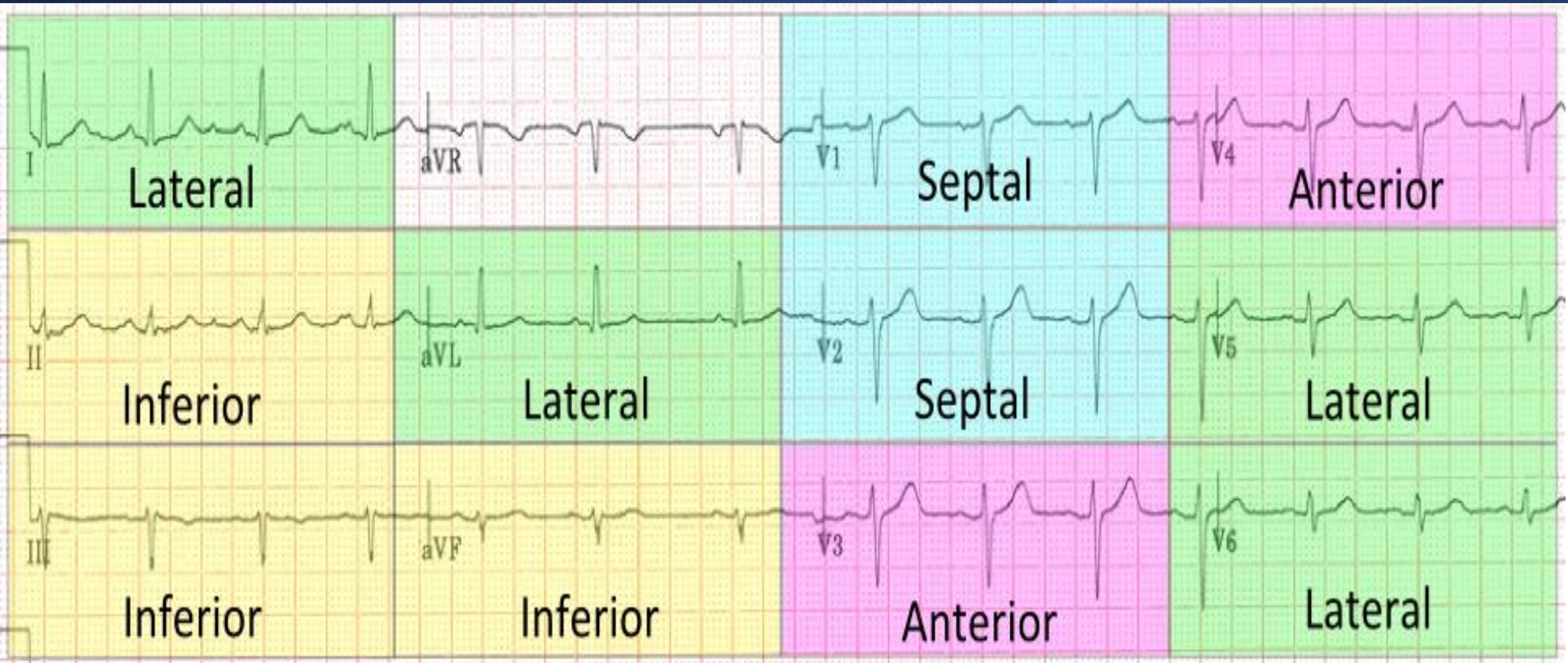
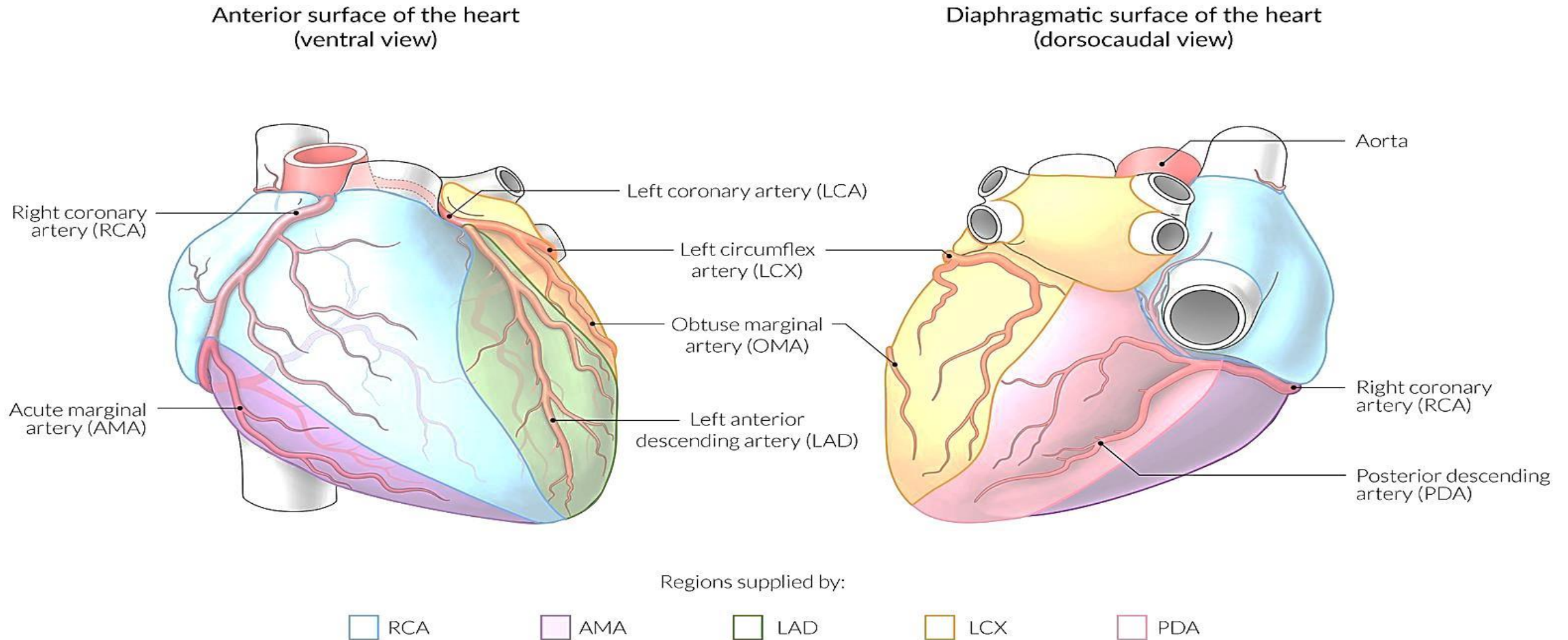


Figure 4.24. ECG evolution during acute ST elevation myocardial infarction (also termed “acute Q wave myocardial infarction”). However, as described in Chapter 7, if successful early reperfusion of the coronary occlusion is achieved, the elevated ST segments return to baseline without subsequent T wave inversion or Q wave development.

ECG Changes in STEMI



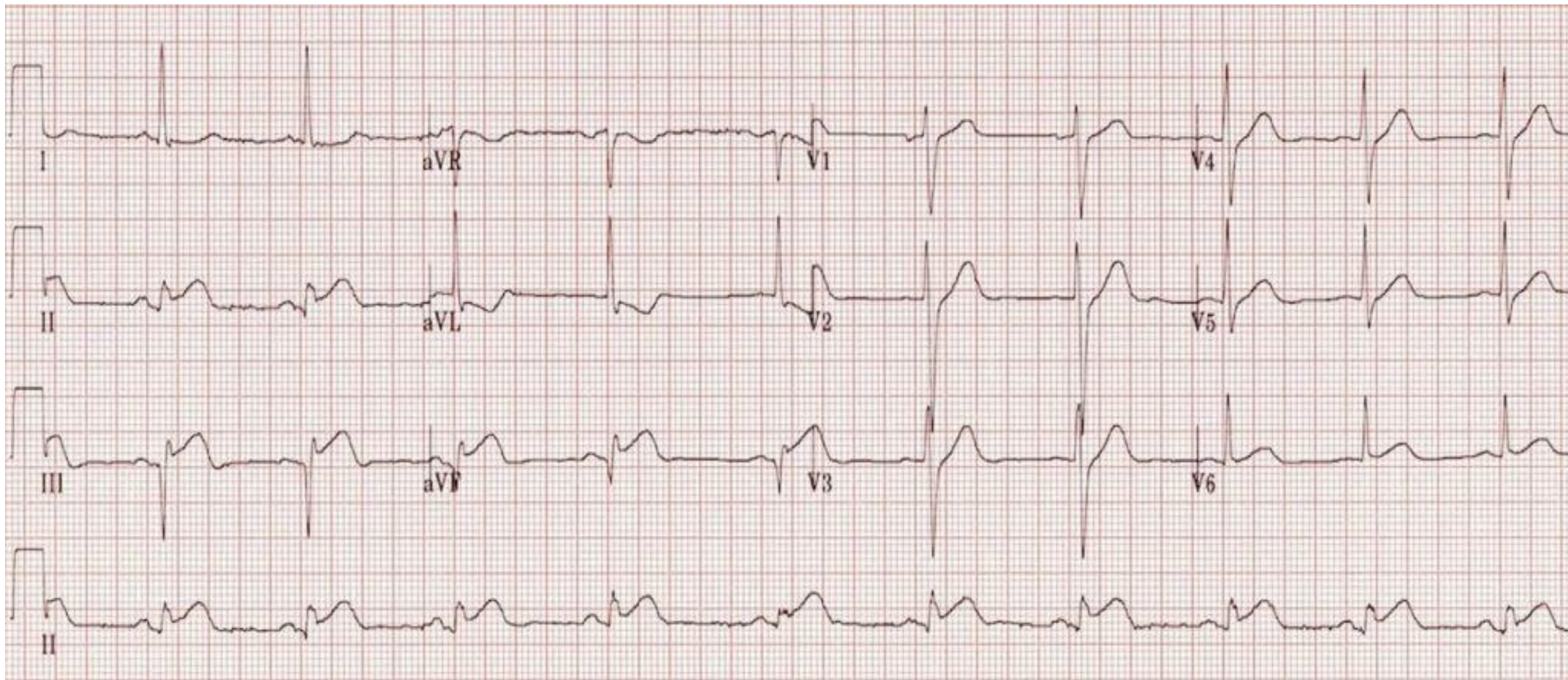
ECG Changes in STEMI



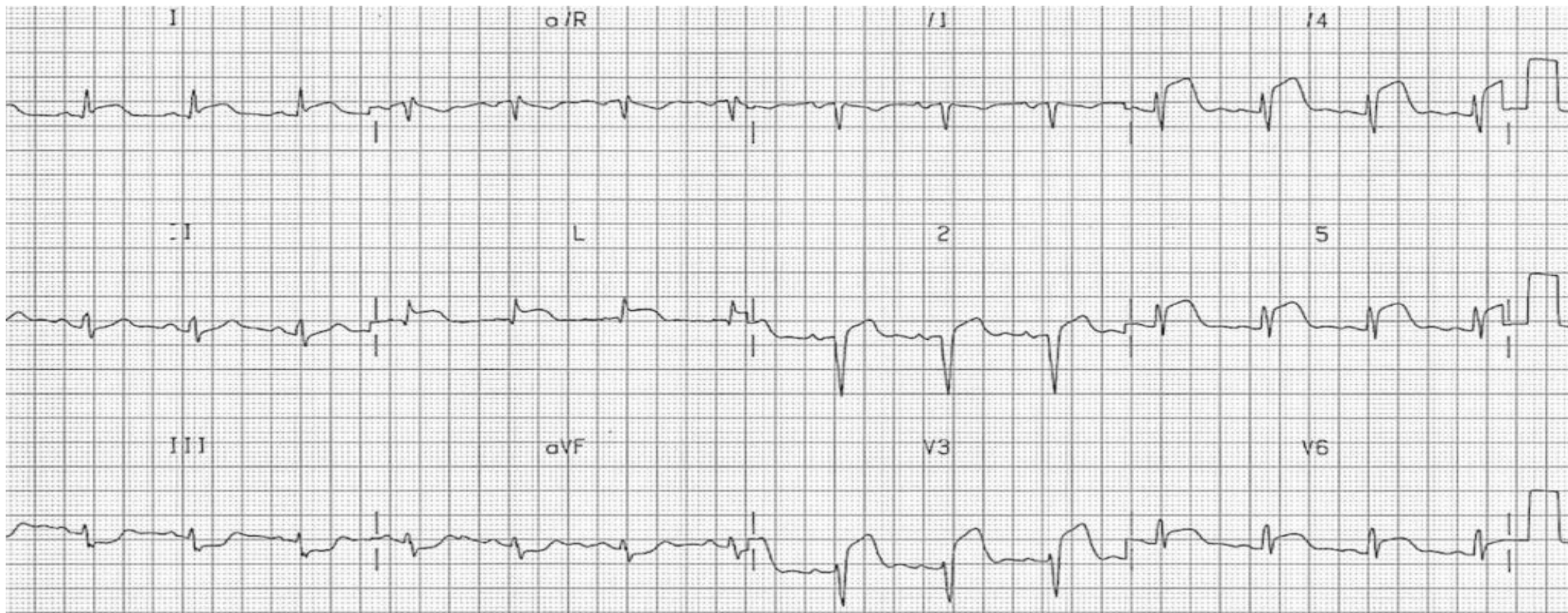
ECG Changes in STEMI

ECG Localization of STEMI	Infarct Location	Leads with ST Elevation or Q Waves	Vessel Involved
	AnteroSeptal	V1-V2	Left Anterior Descending (LAD)
	AnteroApical	V3-V4	Distal LAD
	AnteroLateral	V5-V6	Left Circumflex (LCX)
	Lateral	I, aVL	LCX
	Inferior	II, III, aVF	Right Coronary artery (RCA)
	Posterior	V7-V9	Posterior descending artery (PDA)

ECG Changes in STEMI



ECG Changes in STEMI



Management - STEMI

- Oxygen ($\text{SpO}_2 < 90\%$)
- Revascularization
- Medications

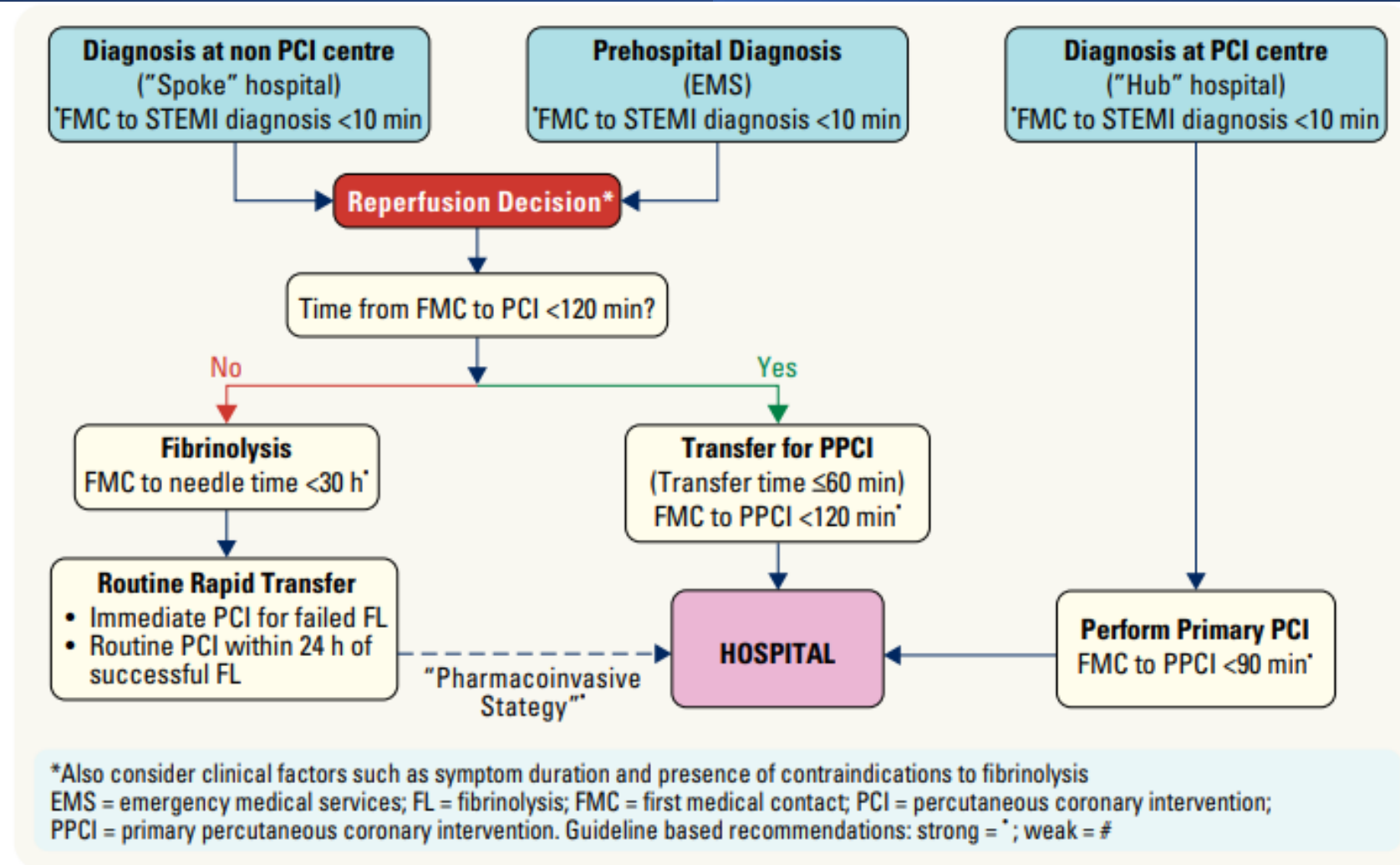


Figure 36. Reperfusion strategy in STEMI

Management - STEMI

Prompt coronary reperfusion restores blood flow, limits myocardial damage and reduces mortality

Emergency coronary angiography with PCI

- **Indication:** Within 12 hours of symptom onset AND
- Within 90 minutes from first medical contact at PCI capable facility OR
- Within 120 minutes from first medical contact at non-PCI capable facility
- **Procedure:** balloon dilatation with stent implantation

Fibrinolytic therapy in STEMI

- **Indications:**
- PCI cannot be performed ≤ 120 minutes from contact with medical provider
- No contraindications to fibrinolysis present
- **Agents:** Tenecteplase, Alteplase, Reteplase, Streptokinase

Management - STEMI

Table 10. Contraindications for Thrombolysis in STEMI

Absolute	Relative
Prior intracranial hemorrhage	Chronic, severe, poorly controlled HTN
Known structural cerebral vascular lesion	Uncontrolled HTN (sBP >180, dBP >110)
Known malignant intracranial neoplasm	Current anticoagulation
Significant closed-head or facial trauma (≤ 3 mo)	Noncompressible vascular punctures
Ischemic stroke (≤ 3 mo)	Ischemic stroke (≥ 3 mo)
Active bleeding	Recent internal bleeding (≤ 2 -4 wk)
Suspected aortic dissection	Prolonged CPR or major surgery (≤ 3 wk)
	Pregnancy
	Active peptic ulcer disease

Management - STEMI

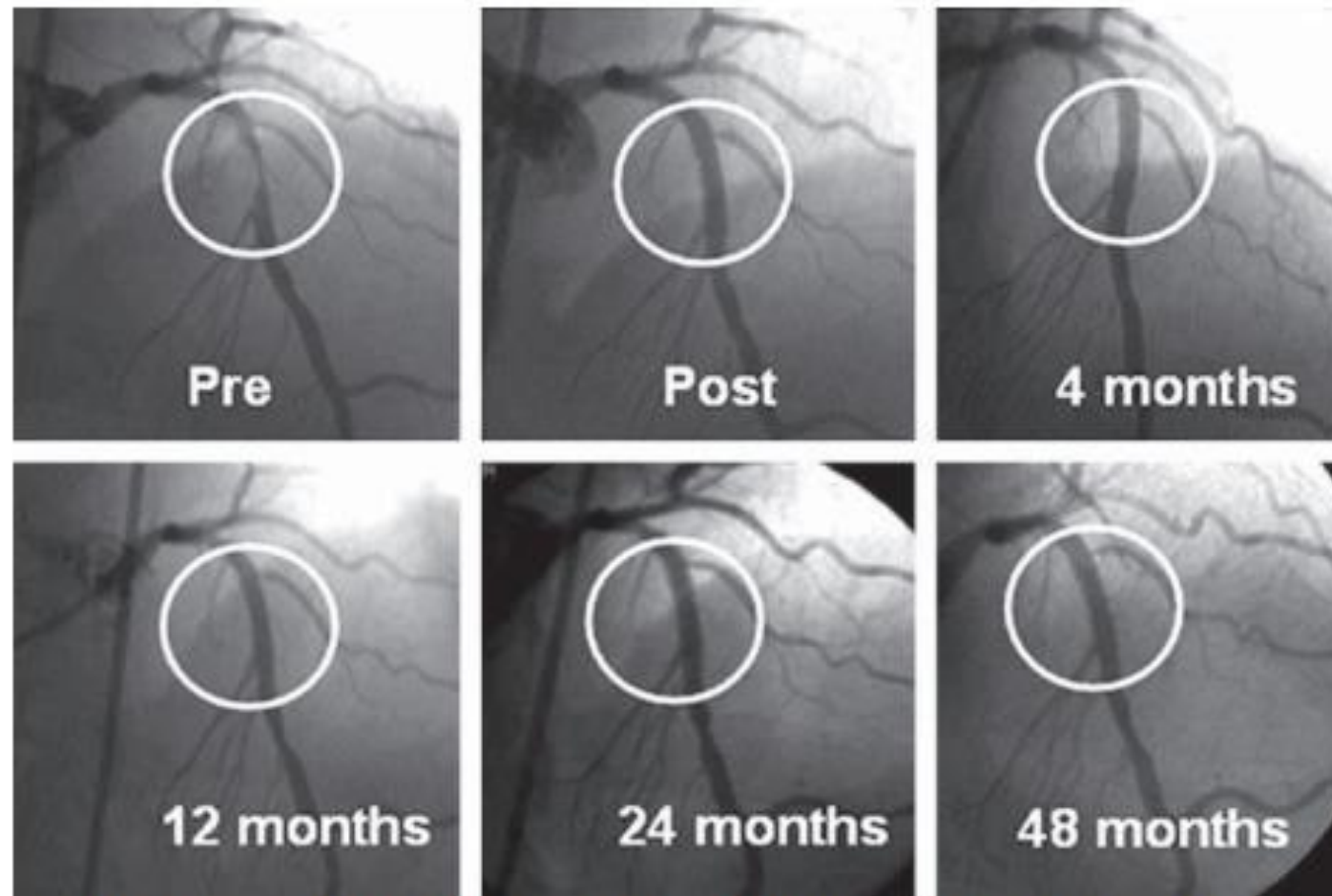


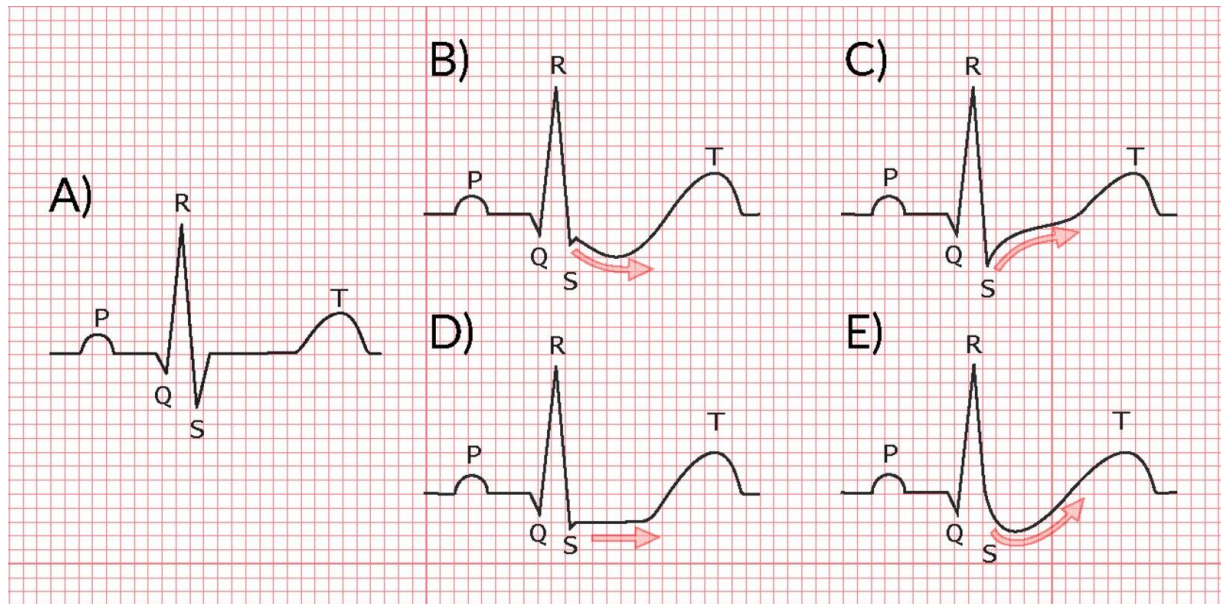
FIGURE 270-3 Long-term results from one of the first patients to receive a sirolimus-eluting stent from early Sao Paulo experience. (From GW Stone, in D Baim [ed]: *Cardiac Catheterization, Angiography and Intervention*, 7th ed. Philadelphia, Lippincott Williams & Wilkins, 2006; with permission.)

Management - STEMI

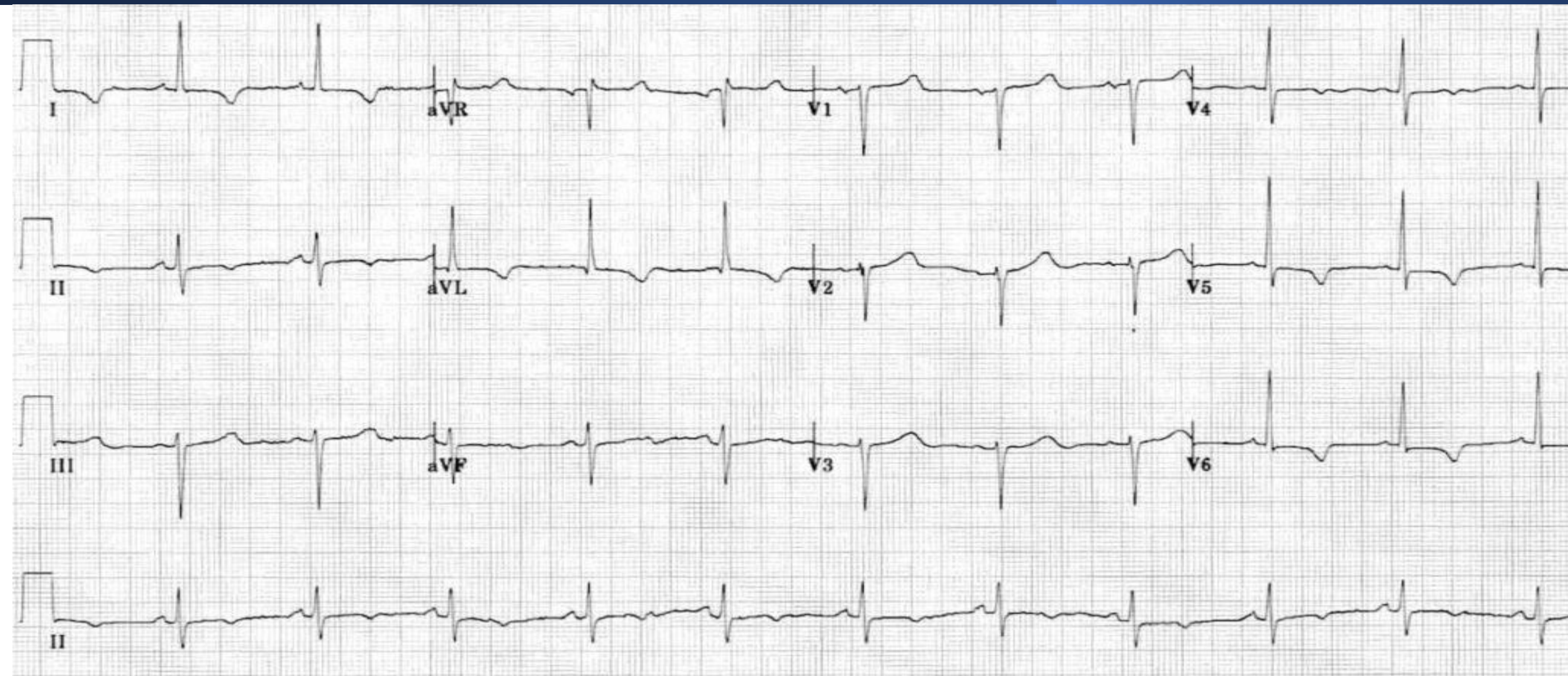
- Morphine
- Oxygen ($\text{SpO}_2 < 90\%$)
- Nitrates
- ASA
- Beta blockers (only if not in HF or at risk for shock)
- ACEi (STEMI, HTN, DM, & HFrEF)
- Statin (High intensity)
- Heparin (48 Hours)
- Clopidogrel (P2Y₁₂ receptor blocker)

ECG Changes in NSTEMI/UA

- ECG changes in NSTEMI/UA
- **No ST elevations present**
- Nonspecific signs of ischemia may be present (ST depression, & T-wave inversions)



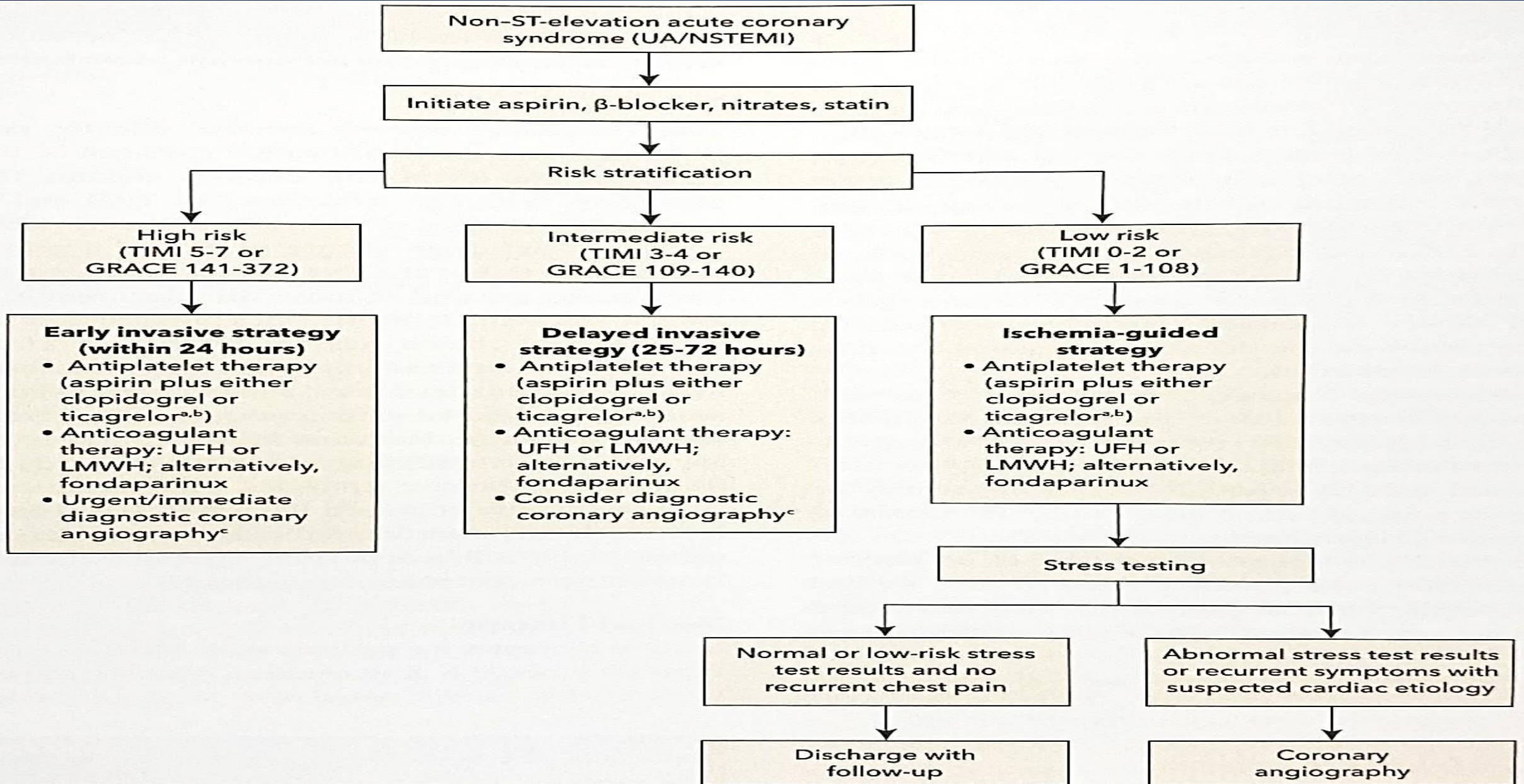
ECG Changes in NSTEMI/UA



Management – NSTEMI/UA

- Morphine
- Oxygen ($\text{SpO}_2 < 90\%$)
- Nitrates
- ASA
- Beta blockers
(only if not in HF or at risk for shock)
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Management – NSTEMI/UA



Management – NSTEMI/UA

- More than 90% of patients improve with the above medical regimen within 1 to 2 days
- Further management depends on mortality risk (e.g., TIMI score), clinical findings, and the availability of resources.
- Invasive Versus Ischemia-Guided Treatment

The screenshot shows the GRACE ACS Risk Model calculator. It has two tabs: 'At Admission (in-hospital/to 6 months)' and 'At Discharge (to 6 months)'. The 'At Admission' tab is active. It contains several input fields: Age (50-59), HR (50-69), SBP (120-139), Creat. (1.2-1.59), and CHF (I (no CHF)). There are also checkboxes for 'Cardiac arrest at admission', 'ST-segment deviation', and 'Elevated cardiac enzymes/markers'. A table shows the probability of Death and Death or MI for In-hospital and To 6 months. At the bottom, there are buttons for 'SI Units', 'Reset', and 'Display Score'.

Probability of	Death	Death or MI
In-hospital	0.4%	4%
To 6 months	1%	10%

TIMI UA/NSTEMI RISK SCORE

1) Age ≥ 65	1 point
2) ≥ 3 risk factors for CAD	1 point
3) Use of ASA (last 7 days)	1 point
4) Known CAD (prior stenosis $\geq 50\%$)	1 point
5) > 1 episode rest angina in < 24 h	1 point
6) ST-segment deviation	1 point
7) Elevated cardiac markers	1 point

doi:10.1371/journal.pone.0007947.t001

Management – NSTEMI/UA

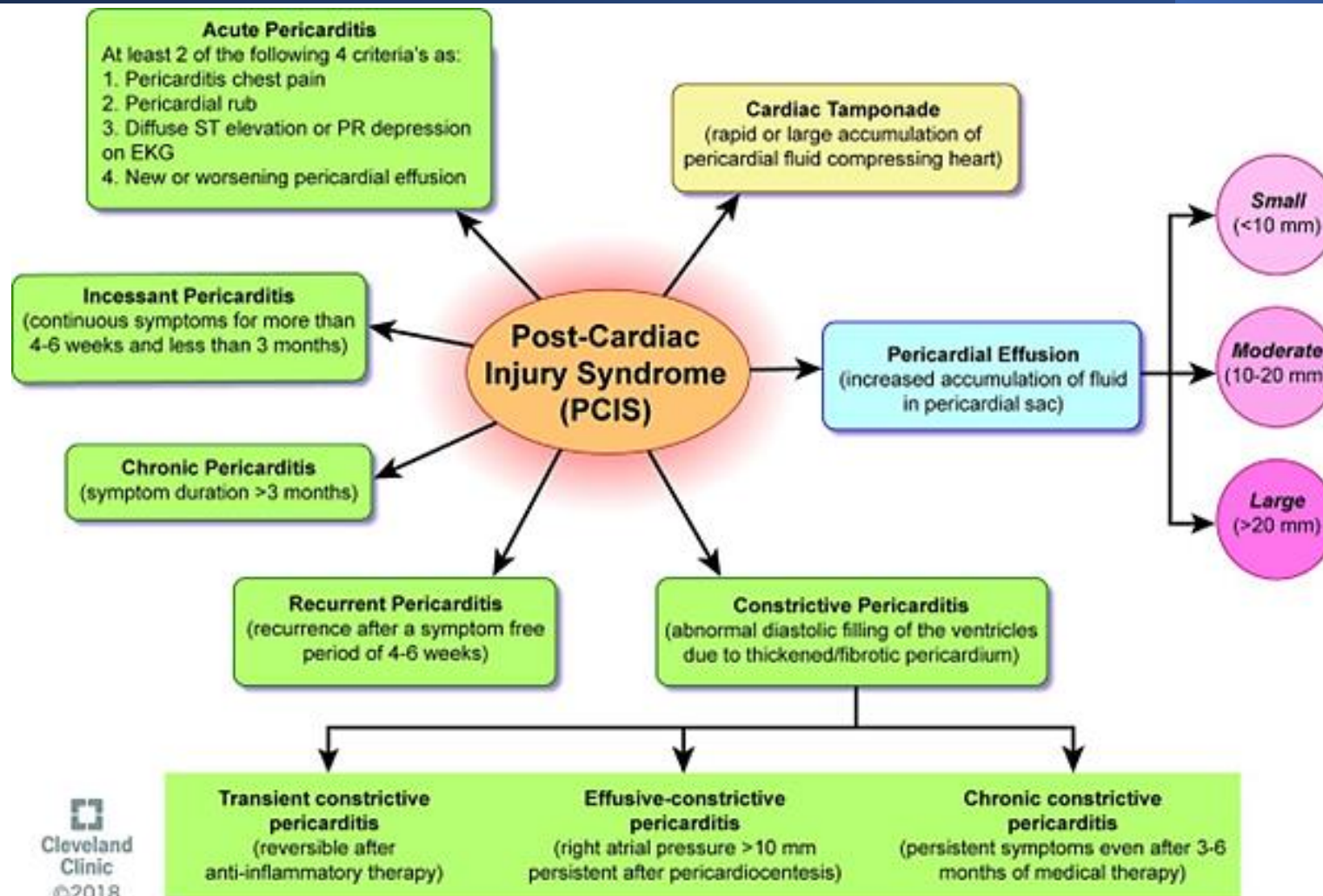
Indications for urgent revascularization in NSTEMI:

- Hemodynamic instability
- Life-threatening arrhythmias
- Refractory ischemic pain
- Acute heart failure
- Mechanical complications (e.g., new/aggravated mitral regurgitation)

MI Complications

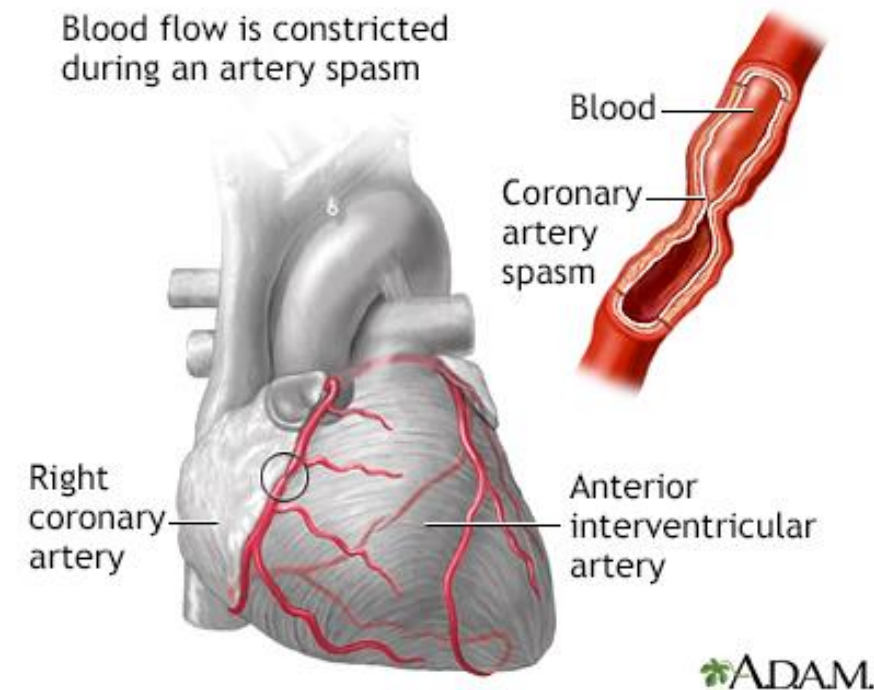
0-24 Hours Post-infarction	• Sudden cardiac death • Arrhythmias (Most common cause of death in MI patients in the first 24 hours) • Acute left heart failure • Cardiogenic shock
1-3 Days Post-infarction	• Acute pericarditis
3-14 Days Post-infarction	• Papillary muscle rupture (MR) • Free wall rupture (Tamponade) • Ventricular septal rupture • Left ventricular pseudoaneurysm
2 weeks to months Post-infarction	• Atrial and ventricular aneurysms • Post myocardial infarction syndrome (Dressler syndrome) • Arrhythmias • HF

MI Complications (post cardiac injury syndrome)



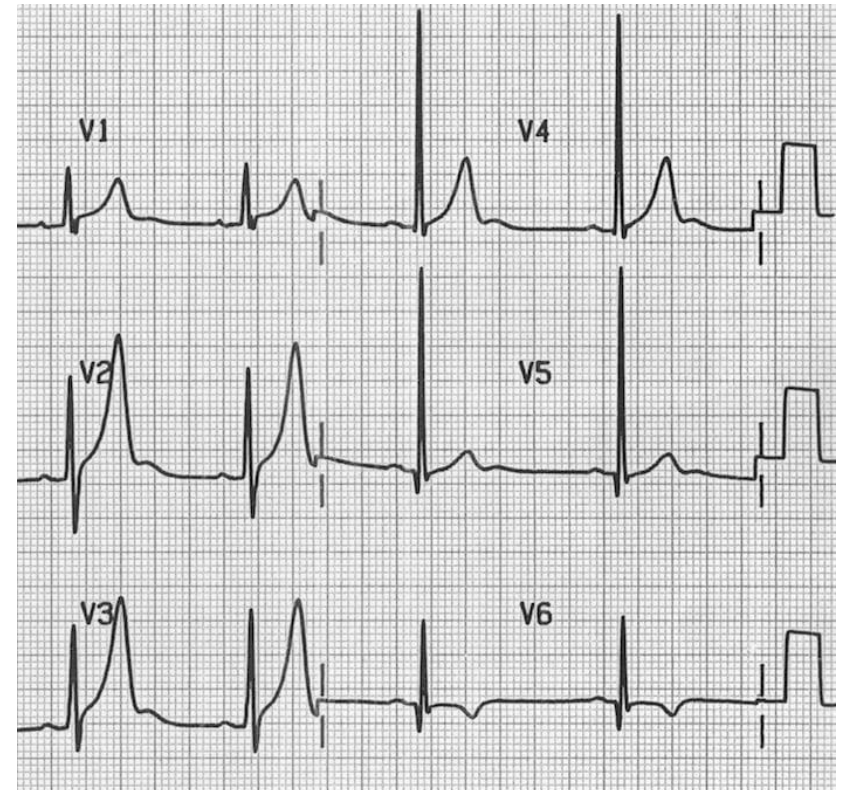
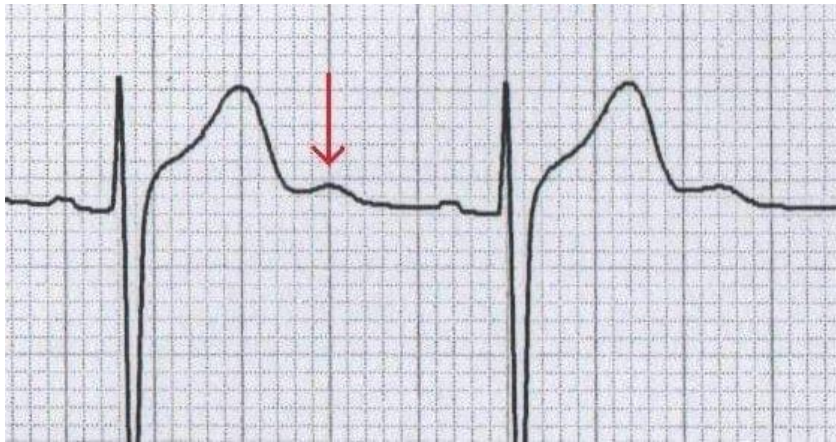
Vasospastic Angina – Definition

- Caused by transient coronary spasms
- Seen often in young women, age of onset ~50 years
- Not affected by exertion (may also occur at rest)
- **Cigarette smoking**; use of stimulants (e.g., **cocaine, amphetamines**), alcohol, or triptans
- Stress, hyperventilation, exposure to cold
- Associated with other vasospastic disorders (e.g., Raynaud phenomenon, migraine headaches)
- Common atherosclerotic risk factors (except smoking) do not apply to vasospastic angina



Vasospastic Angina – Diagnostics

- Resting ECG (Transient ST elevation, U waves)
- Measure serial troponin I and/or troponin T levels during periods of acute chest pain
- Serial troponin levels are unlikely to be elevated in patients with transient ischemic changes

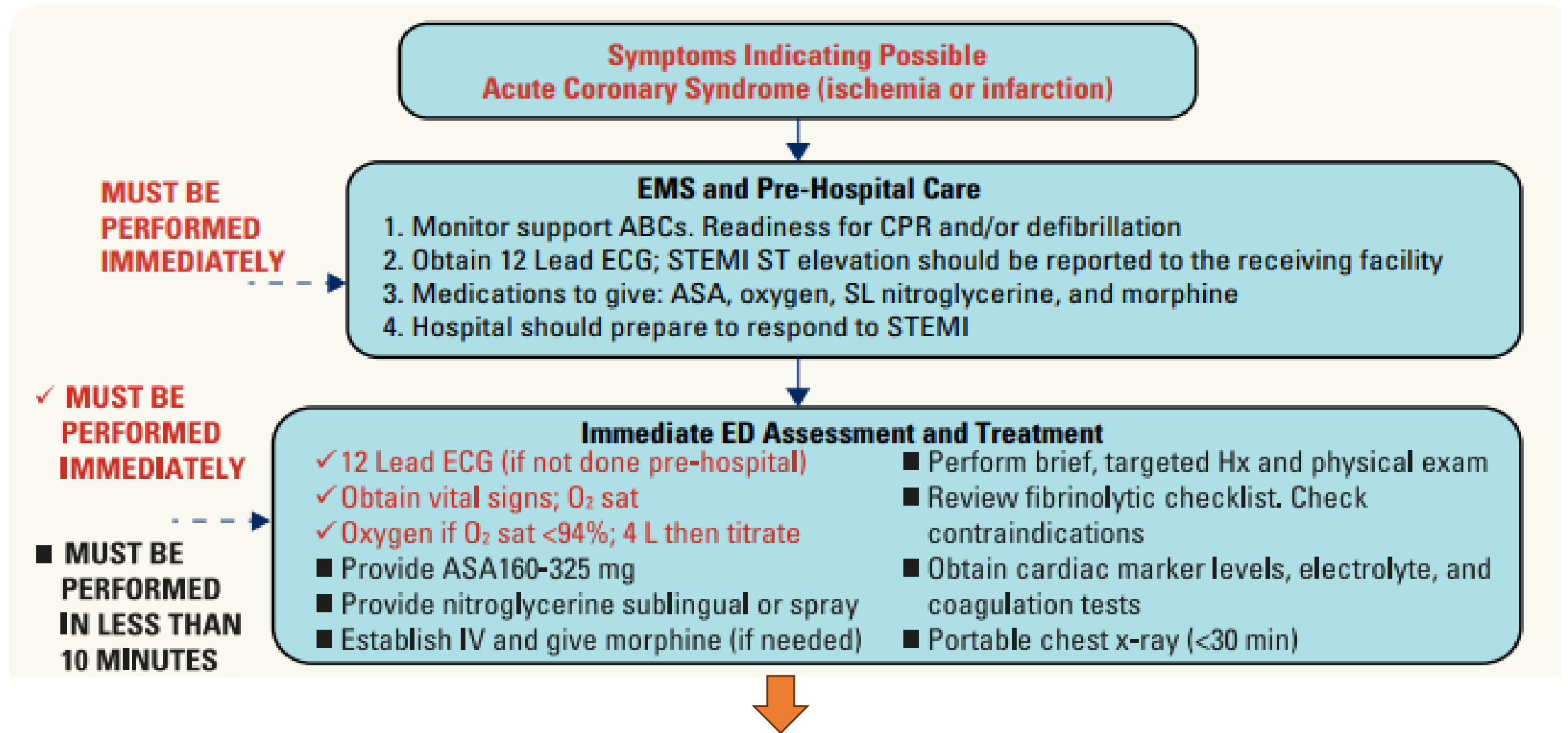


Vasospastic Angina – Management

- **General recommendations**
- Smoking cessation
- **Avoid beta-blockers** and other agents that induce vasoconstriction
- Atherosclerotic risk factor modification, as appropriate
- First-line therapy: **calcium channel blockers**, e.g., verapamil, diltiazem, or nifedipine
- Alternatively: Long-acting nitrates: e.g., isosorbide dinitrate
- Prognosis is good: 5 Year Survival is > 90% with treatment

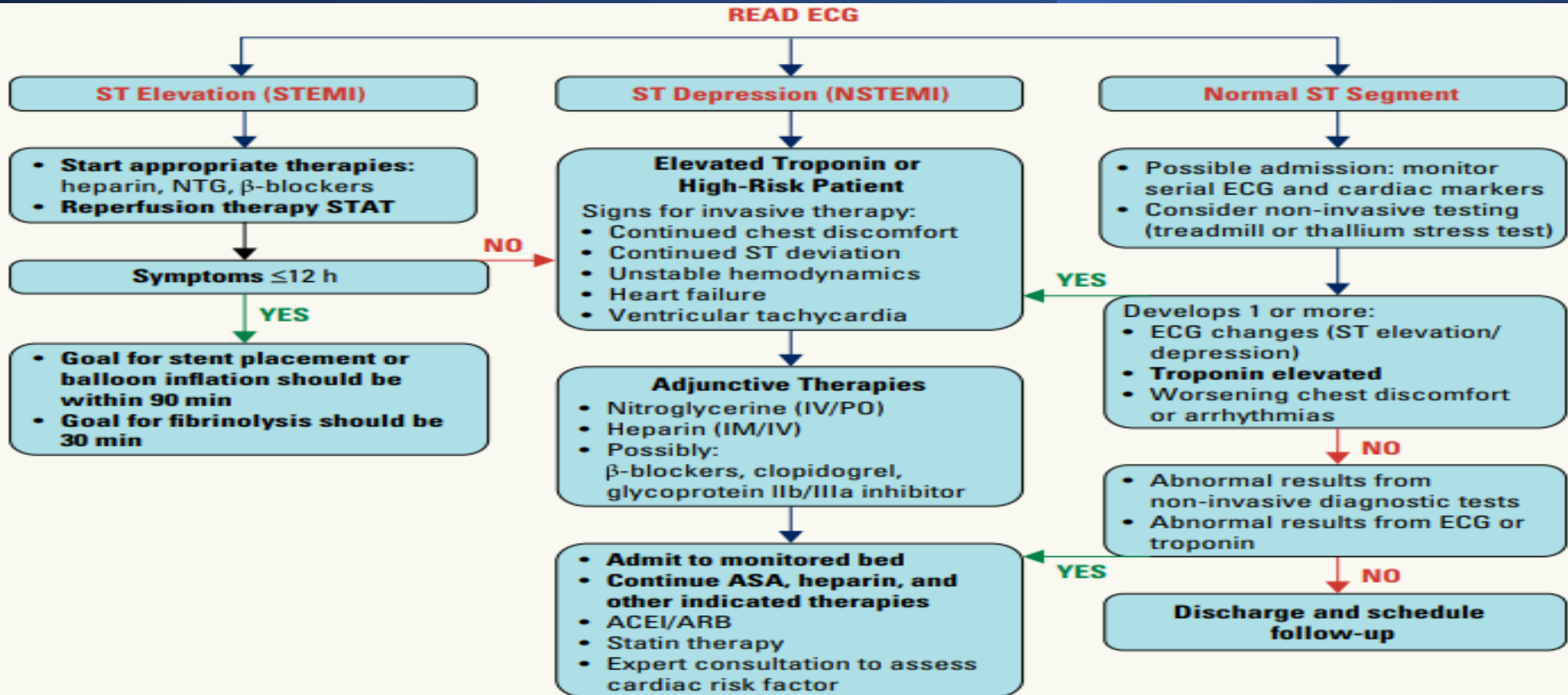
Summary

Treatment Algorithm for Acute Coronary Syndrome



Summary

Treatment Algorithm for Acute Coronary Syndrome



Case Scenario 1

A 54-year-old man comes to the physician because of a 1-week history of chest pain. The pain is episodic and last for 30-45 minutes before it subsided spontaneously; it is localized to the left side of the chest and does not radiate. The patient reports that the pain sometimes occurs when he is angry or upset; he has not noticed any relieving factors. He has two herniated lumbar spinal discs, for which he takes ketorolac occasionally, but is otherwise healthy. He goes swimming 3 times a week and tries to maintain a healthy diet. He drinks 1-2 glasses of wine on weekends. He does not smoke or use illicit drugs. His vital signs are within normal limits. Cardiac examination shows the point of maximal impulse at the 5th intercostal space at the midclavicular line. Auscultation shows no murmurs, rubs, or gallops. An ECG shows a normal sinus rhythm. Which of the following is the most appropriate next step in the evaluation for coronary artery disease in this patient?

- a. No testing needed
- b. Troponin I measurement
- c. Stress myocardial perfusion imaging
- d. Cardiac catheterization
- e. Dipyridamole stress echocardiography
- f. Exercise Stress test

Case Scenario 2

A 72-year-old man comes to the physician because of a 2-month history of intermittent retrosternal chest pain and tightness on exertion. He has type 2 diabetes mellitus, osteoarthritis of the right hip, and hypertension. Current medications include insulin, ibuprofen, enalapril, and hydrochlorothiazide. Vital signs are within normal limits. His troponin level is within the reference range. An ECG at rest shows a right bundle branch block and infrequent premature ventricular contractions. The patient's symptoms are reproduced during adenosine stress testing. Repeat ECG during stress testing shows new ST depression of > 1 mm in leads V2, V3, and V4. Which of the following is the most important underlying mechanism of this patient's ECG changes?

- a. Vasospasm of the coronary arteries
- b. Ruptured cholesterol plaque within a coronary vessel
- c. Diversion of blood flow from stenotic coronary arteries
- d. Transient atrioventricular nodal blockade
- e. Reduced left ventricular preload
- f. Increased myocardial oxygen demand

Questions?

when your lecturer asks if you have any questions



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References & further reading

- Cecil essentials of medicine, 10th edition
- Harrison's principles of internal medicine 20th edition
- American Heart Association
- Amboss
- Life in the fast lane (LITFL.Com)
- CORE IM
- MKSAP 18

Thank You

one-minute paper (2nd sem. 25-
26)

